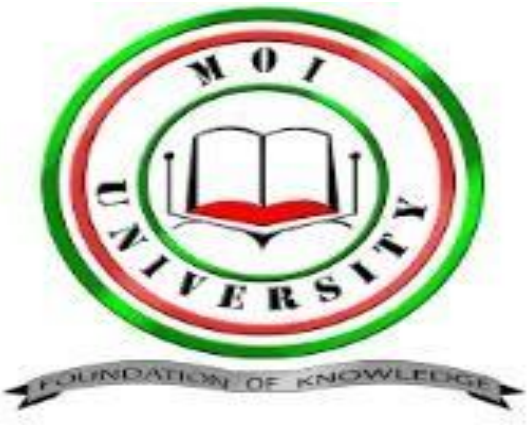


MOI UNIVERSITY

DEPARTMENT OF STATISTICS

A PROJECT ON

STATISTICAL ANALYSIS OF RETAIL SALES PERFORMANCE AND PAYMENT BEHAVIOR ACROSS STORES AND TIME

Submitted by

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Under the Guidance of

DR. EDNA CHUMO
Supervisor, Department of Statistics

For Partial Fulfilment of
Applied Statistics with Computing Course
Academic Year 2025/2026

DECLARATION

We declare that this project titled "Statistical Analysis of Retail Sales Performance and Payment Behavior Across Stores and Time" is our original work and has not been submitted to any other university or institution for the award of any academic qualification. All sources of information used in this study have been duly acknowledged and referenced.

This project has been carried out under the guidance and supervision of Dr. Edna Chumo in partial fulfilment of the requirements for the course Applied Statistics with Computing for the Academic Year 2025/2026.

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This is to certify that the project titled:

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ABSTRACT

This project applies statistical methods to analyze real-world retail transaction data order to understand sales performance patterns and consumer payment behavior across different stores and time periods. Using a structured retail transactions dataset of 2,000 observations covering multiple stores over 2023–2025, the study explores patterns and relationships within the data.

The analysis employs descriptive statistics to summarize sales trends, inferential statistical methods including chi-square tests, t-tests, and analysis of variance (ANOVA) to compare sales performance across stores, products, and time periods. Regression analysis identifies factors influencing total sales value, while time-based analysis examines variations in sales across days of the week and times of the day. All analyses were conducted using R statistical software.

Key findings show that Quantity and Unit Price are the dominant drivers of total sales value ($R^2 = 0.936$), while store identity, location, and payment type have no statistically significant independent effect on sales. Monthly seasonality significantly influences sales patterns ($p = 0.006$), and time of day produces statistically distinct sales outcomes ($p = 0.049$). The SARIMA seasonal time series model provides the best forecasting fit with MAPE = 10.72%.

These findings provide actionable insights into sales optimization, customer payment preferences, and evidence-based retail management strategies.

Keywords: Retail sales, descriptive statistics, ANOVA, regression analysis, time series, payment behavior, R software

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LIST OF ABREVIATIONS

Statistical Methods and Tests	
ANOVA	Analysis of Variance
MANOVA	Multivariate Analysis of Variance
ARIMA	Autoregressive Integrated Moving Average
SARIMA	Seasonal Autoregressive Integrated Moving Average
SARIMAX	Seasonal Autoregressive Integrated Moving Average with Exogenous Regressors
ACF	Autocorrelation Function
PACF	Partial Autocorrelation Function
HSD	Honestly Significant Difference (Tukey HSD)
VIF	Variance Inflation Factor
AIC	Akaike Information Criterion
MAPE	Mean Absolute Percentage Error
RMSE	Root Mean Square Error

AUC	Area Under the Curve
ADF	Augmented Dickey-Fuller (test)

Statistical Measures	
R^2	R-squared (coefficient of determination)
SD	Standard Deviation
df	Degrees of Freedom
p	Probability value (p-value)
F	F-statistics
W	Wilcoxon / Shapiro-Wilk statistic
OR	Odds Ratio
β / beta	Regression Coefficient
Eta^2	Eta-squared (effect size)
V	Cramer's V (association measure)
r	Pearson correlation coefficient

Software and Programming	
R	R statistical software
STL	Seasonal-Trend decomposition using LOESS
LOESS	Locally Estimated Scatterplot Smoothing
Models	
AR	Autoregressive
MA	Moving Average
SVM	Support Vector Machine
Dataset Variables and Identifiers	
S1–S10	Store identifiers
C1–C5	Cashier identifiers
AST	Applied Statistics (student registration prefix)
B2B	Business to Business
KES	Kenyan Shilling (currency)
General Academic / Research	
M&E	Monitoring and Evaluation
ROI	Return on Investment
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CHAPTER 1: INTRODUCTION

1.1 Background

The retail industry operates in a highly competitive environment where understanding sales patterns and customer behavior is essential for effective decision-making. With the advancement of digital transaction systems, retailers now collect detailed data on sales, products, payment methods, and customer interactions. However, raw data alone is insufficient without proper statistical analysis.

This project addresses the problem of limited analytical use of retail transaction data in evaluating sales performance and payment behavior across different stores and time periods. By applying statistical techniques, the study seeks to identify significant factors influencing sales and customer payment choices. The significance of this project lies in its ability to transform transactional data into meaningful insights that can assist retail managers in improving operational efficiency, optimizing product offerings, and enhancing customer experience.

1.2 Problem Statement

Despite the availability of detailed retail transaction data, organizations often focus mainly on recording transactions rather than performing comprehensive statistical analysis. As a result, valuable insights related to sales trends, payment preferences, and time-based performance variations remain unexplored. This creates a need for systematic analytical approaches that transform raw transaction data into meaningful, actionable information for decision-makers.

1.3 Objectives

1.3.1 General Objective

To statistically analyze retail sales performance and payment behavior across stores and time using descriptive, inferential, and regression-based statistical methods.

1.3.2 Specific Objectives

1. To statistically describe and compare sales patterns across products, stores, and time periods.
2. To statistically examine the association between payment methods and product categories.
3. To statistically compare mean sales performance across stores, locations, and time of day.
4. To determine statistically significant factors influencing total sales value using regression modelling.
5. To assess temporal patterns and variations in retail sales performance across dates, monthly periods, hours, days of the week, and times of the day using time-based statistical analysis.

1.4 Significance of the Study

This study makes a practical contribution to retail management by demonstrating how statistical analysis can convert transaction records into operational insights. Findings on peak sales periods, payment preferences, and revenue drivers directly inform staffing, inventory, and marketing decisions. The project also provides a replicable analytical framework applicable to other retail environments and serves as a practical platform for applying statistical concepts from the Applied Statistics with Computing course.

1.5 Project Timeline

Activity	Duration	Deliverable
Proposal for development and approval	2	Approved proposal
Data preparation and cleaning	2	Clean dataset
Descriptive data analysis	2	Summary statistics and charts
Inferential analysis and modeling	3	Statistical test results
Interpretation of results	2	Analytical conclusions
Report writing	3	Draft project report
Final review and submission	2	Final project submission
Total Duration	**16 weeks**	**Completed project**

Table 1.5: Project Timeline

1.6 Resources Required

- Retail transactions dataset (2,000 observations, 14 variables)
- Computer or laptop
- Statistical software: R (free, open source)
- Internet access for research and references
- Textbooks and lecture notes on statistical methods
- Supervision and guidance from Dr. Edna Chumo

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing knowledge on retail sales analysis and payment behavior to understand current analytical practices and identify gaps that align with the objectives of this project.

2.2 Retail Transaction Data and Sales Analysis

Retail businesses generate large volumes of transactional data through daily sales activities, capturing important operational details such as products purchased, quantities sold, prices, payment methods, store locations, and timing of purchases. With the advancement of digital sales systems, retailers now possess structured datasets capable of supporting statistical analysis and evidence-based decision-making.

Understanding sales performance and customer payment behavior has become essential for improving operational efficiency and competitiveness in the retail sector. Statistical analysis provides tools for summarizing sales data, identifying patterns, comparing performance across stores, and evaluating customer purchasing behavior over time.

2.3 Current Analytical Practices

Existing analytical practices in retail environments demonstrate that sales performance can be evaluated using measurable indicators such as revenue, quantity sold, and pricing patterns. Analysis of retail transactions generally shows that:

- Sales performance varies across stores and locations due to operational and environmental differences.
- Product categories contribute differently to overall revenue depending on demand patterns.
- Customer payment methods differ based on transaction value and purchasing conditions.
- Sales activity changes across times of the day and days of the week, indicating temporal trends.
- Statistical tools such as descriptive summaries, hypothesis testing, and regression models can evaluate relationships among sales variables.

Retail data analysis also highlights the importance of combining multiple variables simultaneously rather than examining them independently. Integrating product, payment, and time-related information provides a more comprehensive understanding of business performance.

2.4 Identified Gaps

Although retail transaction systems collect extensive data, several gaps remain in how such data is utilized for analytical and decision-making purposes:

- Limited Analytical Use of Transaction Data: Retail datasets are often used primarily for accounting and reporting rather than detailed statistical analysis.
- Separation of Sales and Payment Analysis: Sales performance and payment behavior are frequently evaluated independently, despite their potential relationship.
- Insufficient Statistical Comparison Across Stores: Differences in store performance are commonly observed but not statistically tested.
- Underutilization of Time-Based Variables: Variables such as time of day, day of week, and monthly trends are recorded but rarely analyzed systematically.
- Lack of Integrated Modeling Approaches: Many evaluations rely on simple summaries instead of regression or inferential methods.
- Limited Monitoring and Evaluation (M&E) Application: Retail performance monitoring frameworks often lack statistical evaluation mechanisms for continuous improvement.

2.5 Summary

The review indicates that while retail transaction data contains rich information, there is limited comprehensive statistical analysis integrating sales performance, payment behavior, and time-related factors within a single framework. This project addresses these gaps by applying descriptive statistics, inferential analysis, and regression modeling to a retail transactions dataset, providing a practical application of statistical techniques while addressing real-world analytical challenges.

CHAPTER 3: METHODOLOGY

3.1 Research Design

The project follows a quantitative analytical design using a retail store transaction dataset containing 2,000 observations and multiple variables describing sales, payment methods, store characteristics, and time-related factors. The study applies statistical techniques to transform raw transaction data into meaningful analytical insights.

3.2 Data Source

The study utilized secondary data obtained from a structured retail store transaction dataset. The dataset contains 2,000 observations and 14 original variables representing sales transactions recorded across multiple stores and time periods spanning January 2023 to February 2025. No surveys or experiments were required; data was imported directly into R statistical software for analysis.

3.3 Dataset Variables

Variables	Type	Description
Date	Date	Transaction date (**2023-01-01 to 2025-02-28**)
Product	Categorical	Type of product (Chair, Desk, Laptop, Monitor, Phone, Printer, Tablet)
Quantity	Numeric	Number of units sold (1–10)
Unit price	Numeric	Price per product unit (5.01–399.84)
Total price	Numeric	Total transaction value (outcome variable)
Store ID	Categorical	Store identifier (**S1–S10**)
Location	Categorical	Store location (**Store A, B, C, D**)
Store Manager	Categorical	Manager name
Cashier	Categorical	Cashier identifier (**C1–C5**)
Payment Type	Categorical	Cash, Credit card, Debit card, Gift card, online
Time Of Day	Ordinal	Morning / Afternoon / Evening
Day Of Week	Ordinal	Monday through Sunday
Hour	Numeric	Exact hour of purchase (9–21, derived from Time)
Month Factor	Ordinal	Year-Month factor for seasonal analysis (derived)
High value	Binary Factor	1 = Total Price ≥ median; 0 = otherwise (derived)

Table 3.3: Dataset variables

3.4 Analytical Approach

The following statistical methods were applied sequentially:

- ✓ Descriptive statistics: mean, median, standard deviation, frequency tables, histograms, boxplots, bar charts.

- ✓ Chi-square test of independence: to examine association between payment methods and product categories.
- ✓ One-way ANOVA and Kruskal-Wallis tests: to compare mean sales across stores, locations, and time periods.
- ✓ Tukey HSD post-hoc tests: to identify which group pairs differ significantly.
- ✓ Multiple linear regression with Box-Cox transformation: to identify factors influencing total sales value.
- ✓ Interaction regression: to test whether Quantity effects depend on Unit price.
- ✓ Logistic regression: to model the probability of high-value transactions.
- ✓ Time series analysis (ARIMA/SARIMA): to examine temporal trends, seasonality, and forecast future sales.

3.5 Risk Management

Potential Risk	Possible Impact	Mitigation Strategy
Data quality issues	Incorrect analysis results	Thorough data cleaning and validation
Software or technical problems	Delay in analysis	Maintain backups; use free opensource R
Time constraints	Incomplete stages	Strict timeline with weekly monitoring
Misinterpretation of results	Flawed conclusions	Statistical verification; supervisor consultation
Loss of data files	Project delays	Regular cloud and external storage backups

Table 3.5: Risk Management

CHAPTER 4: IMPLEMENTATION

4.1 Data Cleaning and Preprocessing

4.1.1 Introduction

Data cleaning is a critical step in statistical analysis to ensure accuracy, consistency and reliability of results. Raw datasets often contain inconsistencies such as incorrect data types, missing values and improperly formatted date and time variables. This section describes the procedures used to prepare the retail sales transactions dataset for statistical analysis.

4.1.2 Loading the Dataset

```
library(readxl)
Retail_data <- read_excel("Sta450_Final_Project/Retail_data.xlsx")
```

The first 8 rows were inspected to confirm correct data import:

Date	Time	Store ID	Location	Product	Qty	Unit Price	Payment
2024-03-30	15:29	S3	Store C	Tablet	3	364.22	Gift Card
2023-03-16	16:53	S1	Store B	Printer	9	384.74	Online
2024-04-26	21:22	S9	Store B	Laptop	9	397.80	Debit Card
2025-02-08	14:28	S9	Store C	Monitor	7	80.27	Online

4.1.3 Initial Inspection

```
dim(Retail_data)
glimpse(Retail_data)
```

```
## [1] 2000   14
## Rows: 2,000  Columns: 14
```

4.1.4 Missing Values and Duplicates

```
colSums(is.na(Retail_data))    # Check for missing values
sum(duplicated(Retail_data))   # Check for duplicates
```

```
## All 14 variables: 0 missing values
## [1] 0  — No duplicate records found
```

No missing values were detected across all variables, indicating a complete dataset. No duplicate records were found, confirming that each row represents a unique sales transaction.

4.1.5 Data Type Corrections and Factor Ordering

```
Retail_data$date <- as.Date(Retail_data$Date)
```

```
# Convert all character variables to factors
Retail_data <- Retail_data %>%
  mutate(across(where(is.character), as.factor))

# Order DayOfWeek and TimeOfDay as ordered factors
Retail_data$DayOfWeek <- factor(Retail_data$DayOfWeek,
levels =
c("Monday","Tuesday","Wednesday","Thursday","Friday","Saturday","Sunday"),
ordered = TRUE)
Retail_data$TimeOfDay <- factor(Retail_data$TimeOfDay,
levels = c("Morning","Afternoon","Evening"), ordered = TRUE)
```

Ordering temporal categories preserves natural sequencing in statistical models and improves interpretability.

4.1.6 Creating Derived Variables

```
# Binary high-value transaction indicator (for logistic regression)
Retail_data <- Retail_data %>%
  mutate(
    Highvalue = ifelse(TotalPrice >= median(TotalPrice), 1, 0),
    Highvalue = factor(Highvalue, levels = c(0, 1), labels = c("No",
"Yes"))
  )

# Extract hour from Time variable
Retail_data$Hour <- as.numeric(format(Retail_data$Time, "%H"))

# Create MonthFactor for seasonal analysis
Retail_data$Month <- format(Retail_data$Date, "%Y-%m")
Retail_data$MonthFactor <- factor(Retail_data$Month,
levels = sort(unique(Retail_data$Month)), ordered = TRUE)
```

4.1.7 Removing Non-Analytical Variables

```
Retail_data$Time <- NULL
Retail_data$Month <- NULL
Retail_data$TransactionID <- NULL
```

Transaction ID was excluded as it serves only as a unique record identifier and has no analytical value. Time and Month were removed after deriving Hour and Month factor.

4.1.8 Data Quality Checks

```
summary(Retail_data$Quantity) summary(Retail_data$UnitPrice)
summary(Retail_data$TotalPrice)

## Quantity:    Min=1,   Q1=3,   Median=6,    Mean=5.529, Q3=8,   Max=10 ##
UnitPrice:  Min=5.01, Q1=99.68, Median=202.72, Mean=200.40, Max=399.84
## TotalPrice: Min=5.74, Q1=364.87, Median=838.53, Mean=1103.82, Max=3997.40
```

4.1.9 Outlier Assessment

```
boxplot(Retail_data$TotalPrice, main = 'Outlier check -
TotalPrice') z_scores <- scale(Retail_data$TotalPrice)
length(which(abs(z_scores) > 3))
## [1] 12  # 12 observations with |z-score| > 3
```

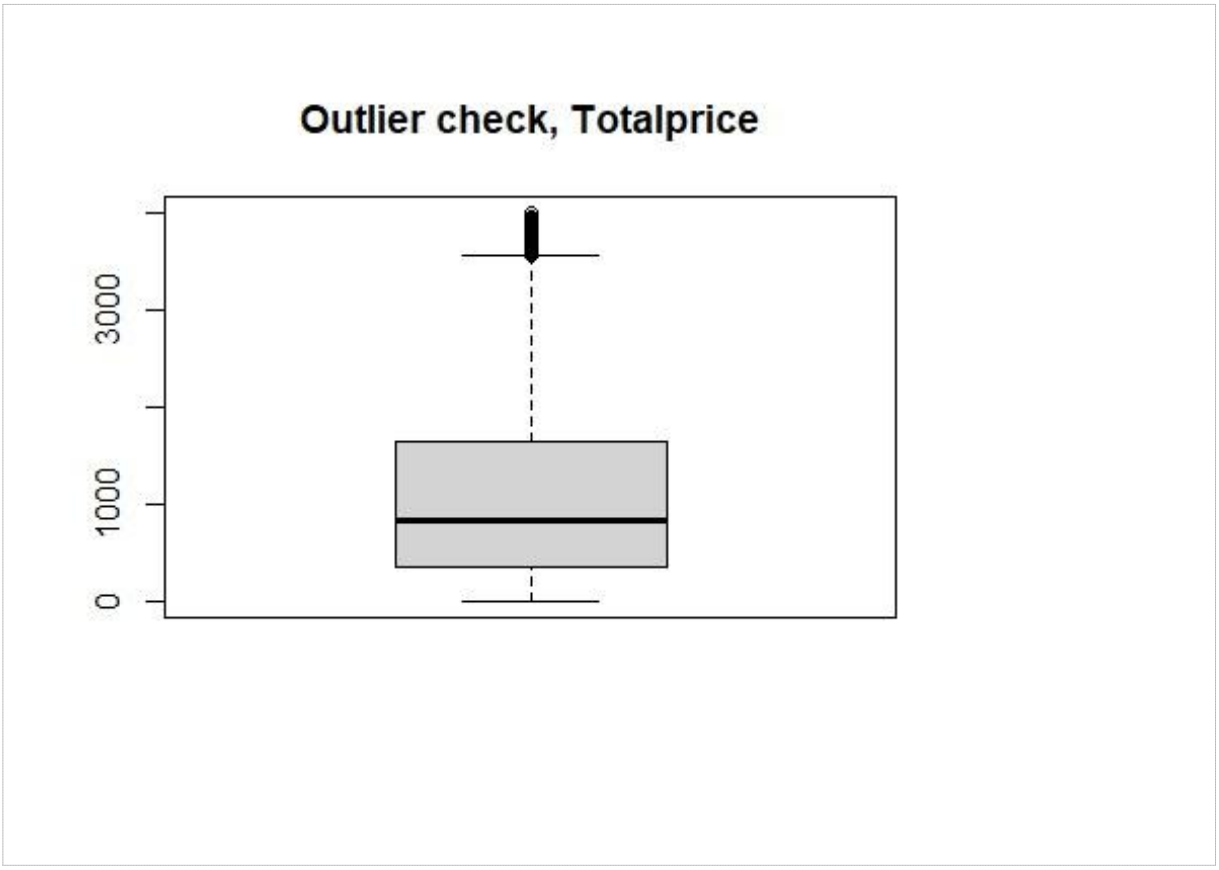


Figure 4.1: Boxplot –Total Price Outlier Check

The boxplot and Z-score analysis identified 12 extreme observations. Given that high value retail transactions are legitimate business events, these records were retained in all analysis.

4.1.10 Final Dataset Structure

```
glimpse(Retail_data)
## Rows: 2,000   Columns: 15
## $ Date        <date>
## $ StoreID     <fct>   S1-S10 ## $
Location    <fct>   Store A-D ## $
Product     <fct>   7 categories
## $ Quantity    <dbl>   1-10 ## $
UnitPrice   <dbl>   5.01-399.84 ## $
PaymentType <fct>   5 categories
## $ Cashier     <fct>   C1-C5
## $ StoreManager<fct>   5 managers
## $ TimeOfDay   <ord>   Morning < Afternoon < Evening
## $ DayOfWeek   <ord>   Monday < ... < Sunday
## $ TotalPrice  <dbl>   5.74-3997.40
## $ Highvalue   <fct>   No / Yes
## $ Hour        <dbl>   9-21
## $ MonthFactor <ord>   2023-01 < ... < 2025-02
```

The cleaned dataset satisfies the assumptions required for regression, ANOVA, chi-square testing, logistic regression, and time series analysis. These steps enhance the reliability, interpretability, and reproducibility of all subsequent statistical analyses.

General Objective: Overview and Descriptive Analysis

Objective: To statistically analyze retail sales performance and payment behavior across stores and time using descriptive, inferential, and regression based statistical methods.

4.2.1 Required Libraries

```
library(dplyr) library(ggplot2)
library(lubridate)
library(knitr)
```

4.2.2 Summary of Numeric Variables

Statistic	Quantity	Unit Price	Price	Hour
Min.	1.000	5.01	5.74	9.00
1st Quartile	3.000	99.68	364.87	12.00
Median	6.000	202.72	838.53	15.00
Mean	5.529	200.40	1103.82	14.96
3rd Quartile	8.000	298.29	1640.90	18.00
Max.	10.000	399.84	3997.40	21.00

Table 4.2.1: Summary of numeric variables

The dataset consists of 2,000 retail transactions with total sales values ranging from KES 5.74 to KES 3,997.40. The average transaction value is approximately KES 1,103.82, while the median value is KES 838.53. The positive gap between mean and median indicates right-skewed distribution, suggesting the presence of relatively higher-value purchases that inflate the overall average. Customer purchase quantities vary between 1 and 10 items per transaction, with an average of about 5–6 items. Unit prices vary widely (5.01–399.84), reflecting differences in product categories and pricing structures.

4.2.3 Top Categories per Variable

```
categorical_cols <-
c("StoreID","Location","Product","PaymentType","Cashier","StoreManager")
top_categories <- lapply(Retail_data[categorical_cols], function(x) {
names(sort(table(x), decreasing = TRUE)[1:3])
}) kable(as.data.frame(top_categories), caption = 'Top 3 Categories
per Variable')
```

Store ID	Location	Product	Payment Type	Cashier	Store Manager
S8	Store C	Printer	Debit Card	C5	Noah
S3	Store B	Monitor	Credit Card	C1	Olivia
S9	Store D	Tablet	Cash	C3	Mia

4.2.4 Average Sales by Store

```
store_sales <- Retail_data |>
group_by(StoreID) |>
  summarise(
    Mean_Sales   = mean(TotalPrice, na.rm = TRUE),
    Median_Sales = median(TotalPrice, na.rm = TRUE),
    Transactions = n(), .groups = 'drop'
  )
knitr::kable(store_sales, caption = 'Summary Statistics of Sales by Store')
```

Store ID	Mean Sales (KES)	Median Sales (KES)	Transactions
S1	1,161.25	930.86	200
S2	1,005.60	723.08	194
S3	1,131.11	829.69	208
S4	1,070.96	783.28	199
S5	1,038.95	820.20	195
S6	1,117.47	835.32	199
S7	1,105.05	833.04	197
S8	1,174.46	899.85	209
S9	1,123.07	908.25	*207*
S10	1,100.58	929.49	192

Table 4.2.2: Average Sales by Store

```
ggplot(store_sales, aes(x = StoreID, y = Mean_Sales)) +   geom_col(fill =
"steelblue") +   labs(title = "Average Sales by Store", x = "Store ID", y =
"Average Total Sales (KES)") +   theme_minimal()
```

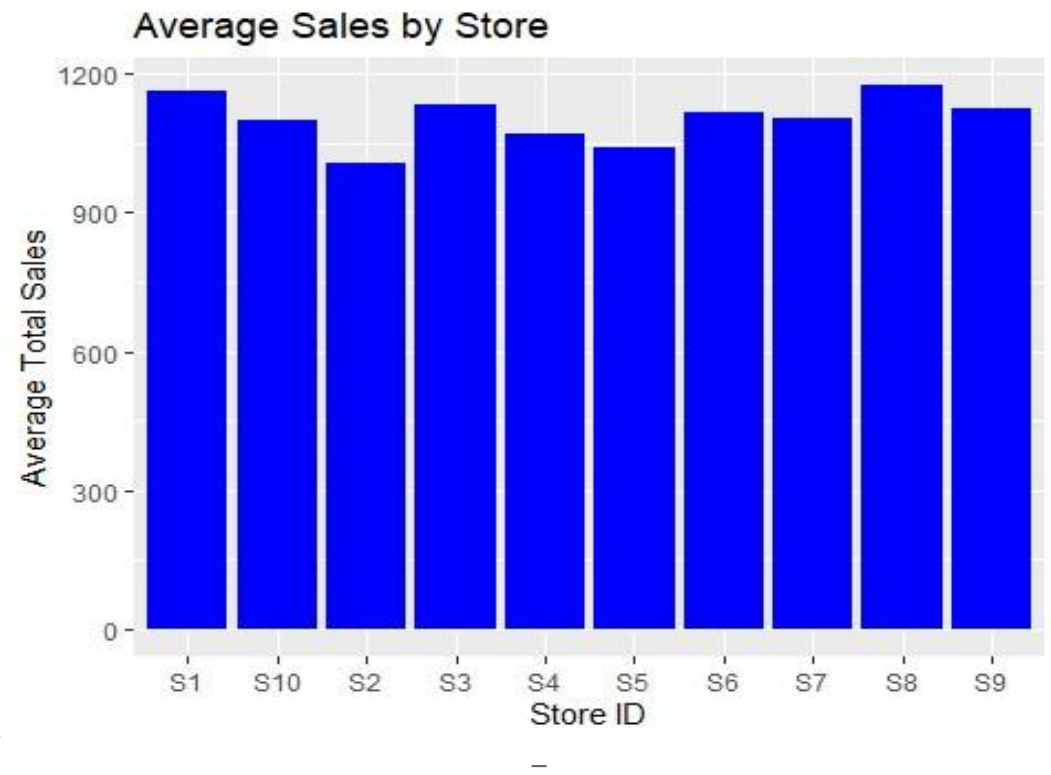


Figure 4.2.1: Bar chart Average sales by Store

Average sales values across stores are relatively similar, ranging from KES 1,006 (S2) to KES 1,174 (S8). Store S8 records the highest average sales, while Store S2 shows comparatively lower average transaction values. The small range of differences suggests fairly consistent customer purchasing behavior across retail locations. This observation is further supported by the ANOVA results which indicate no statistically significant difference in mean sales among stores (F = 0.648, p = 0.756).

4.2.5 Payment Method Distribution

```
payment_summary <- Retail_data %>% count(PaymentType)
ggplot(payment_summary, aes(x = PaymentType, y = n)) +  geom_col(fill =
"steelblue") +  labs(title = "Distribution of Payment Methods", x =
"Payment Type", y =
"Number of Transactions") +
theme_minimal()
```

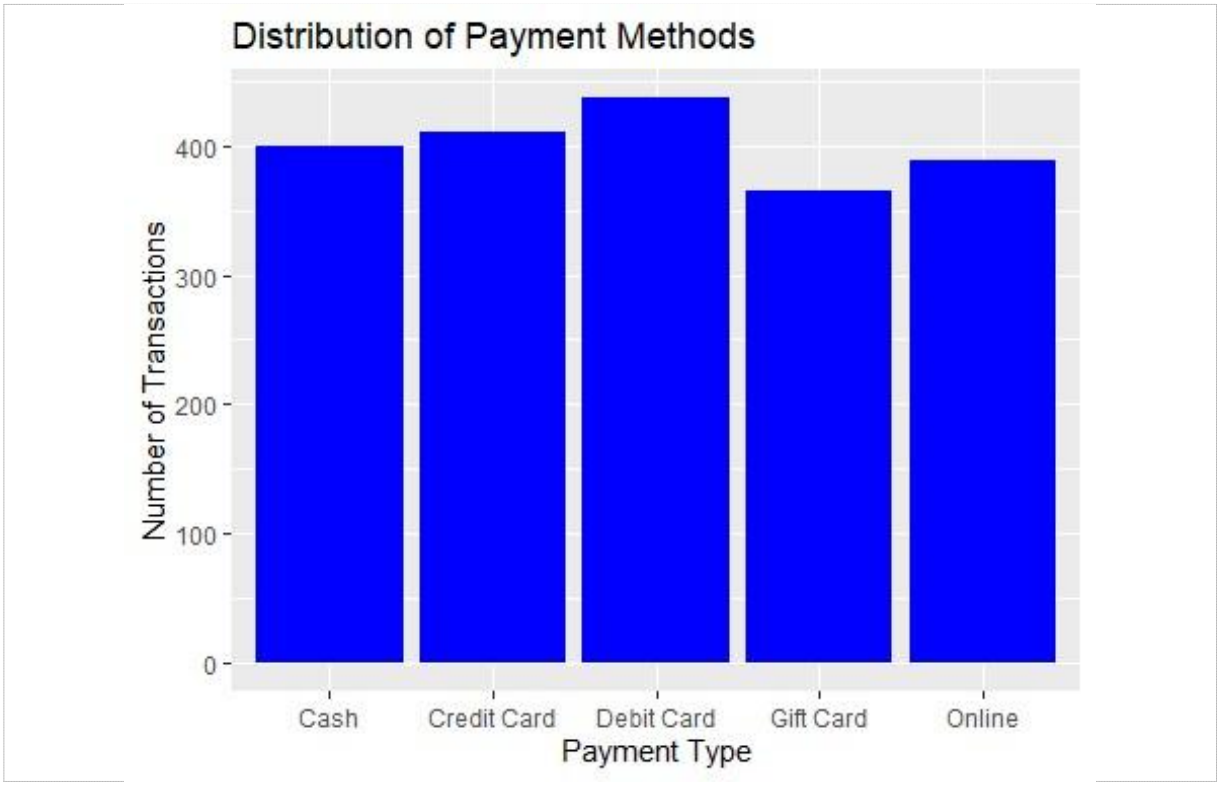


Figure 4.2.2: Bar chart – Distribution of Payment Methods

Payment Type	Transactions	Percentage
Cash	399	20.0%
Credit Card	410	20.5%
Debit Card	437	21.9%
Gift Card	365	18.3%
Online	389	19.5%
Total	2,000	100.0%

The distribution of payment methods is remarkably balanced. Debit Card leads slightly at 21.9%, followed by Credit Card (20.5%), Cash (20.0%), Online (19.5%), and Gift Card (18.3%). This balanced distribution indicates diversified customer payment preferences and reflects effective adoption of multiple payment systems within the retail environment.

4.2.6 Sales Performance Across Time of Day

```
time_sales <- Retail_data %>%  group_by(TimeOfDay) %>%
summarise(MeanSales = mean(TotalPrice, na.rm = TRUE))
time_sales
```

```
## # A tibble: 3 × 2
##   TimeOfDay MeanSales ##
1 Morning   1164.
## 2 Afternoon 1043.
## 3 Evening   1127.
```

```
ggplot(time_sales, aes(x = TimeOfDay, y = MeanSales)) +
geom_col(fill = "steelblue") +  labs(title = "Average Sales by Time
of Day", x = "Time of Day", y =
"Average Sales (KES)") +
theme_minimal()
```

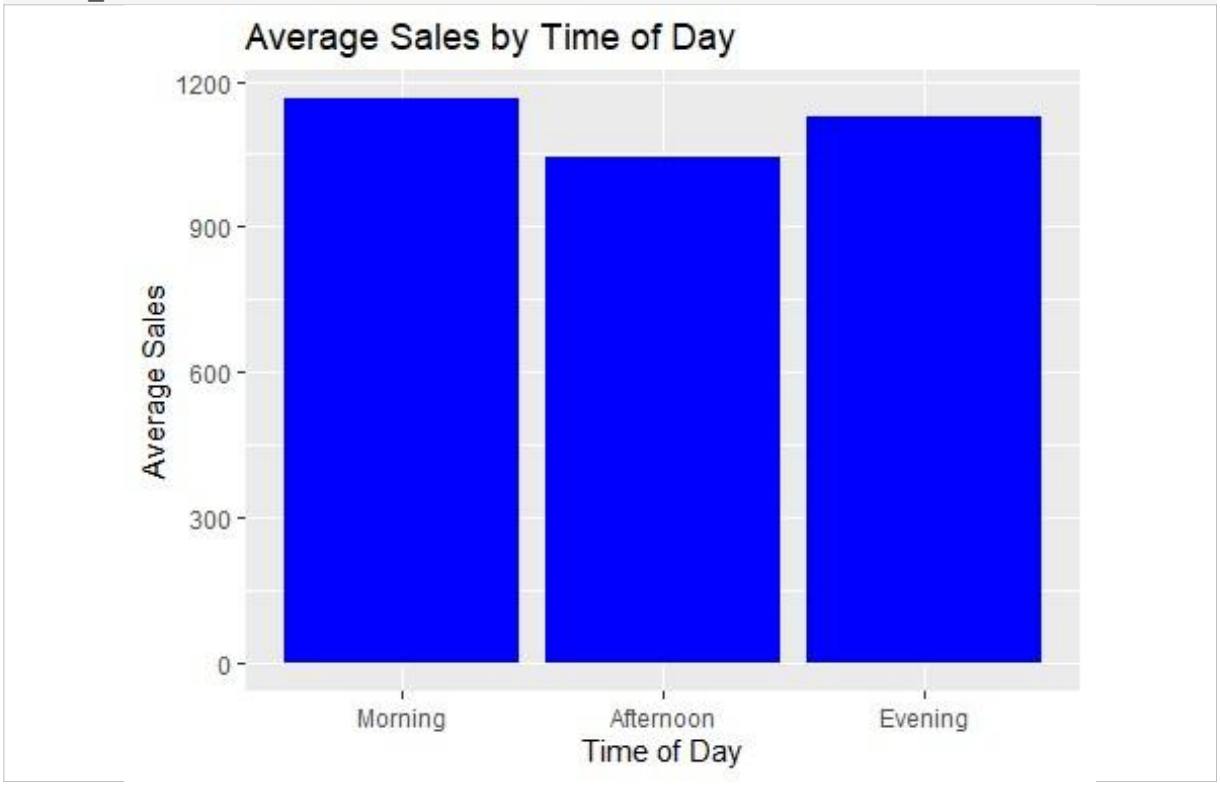


Figure 4.2.3: Bar chart – Average Sales by Time of Day

Morning transactions produce the highest average sales (KES 1,164), followed by Evening (KES 1,127), while Afternoon transactions show slightly lower average spending (KES 1,043). This suggests that customers making purchases earlier in the day tend to spend more per transaction, possibly due to planned shopping behavior. ANOVA confirms a statistically significant difference across time periods (F = 3.016, p = 0.049).

4.2.7 Sales by Day of Week

```
weekly_sales <- Retail_data %>%
group_by(DayOfWeek) %>%
  summarise(MeanSales = mean(TotalPrice, na.rm = TRUE))
ggplot(weekly_sales, aes(x = DayOfWeek, y = MeanSales)) +
geom_col(fill = "steelblue") +  labs(title = "Average Sales by Day of
Week", x = "Day", y = "Average Sales (KES)") +  theme_minimal()
```

```
## DayOfWeek  MeanSales
## Friday     1061. ##
Monday     978.
## Saturday   1042.
## Sunday     1141.
## Thursday   1176.
## Tuesday    1192.
## Wednesday  1135.
```

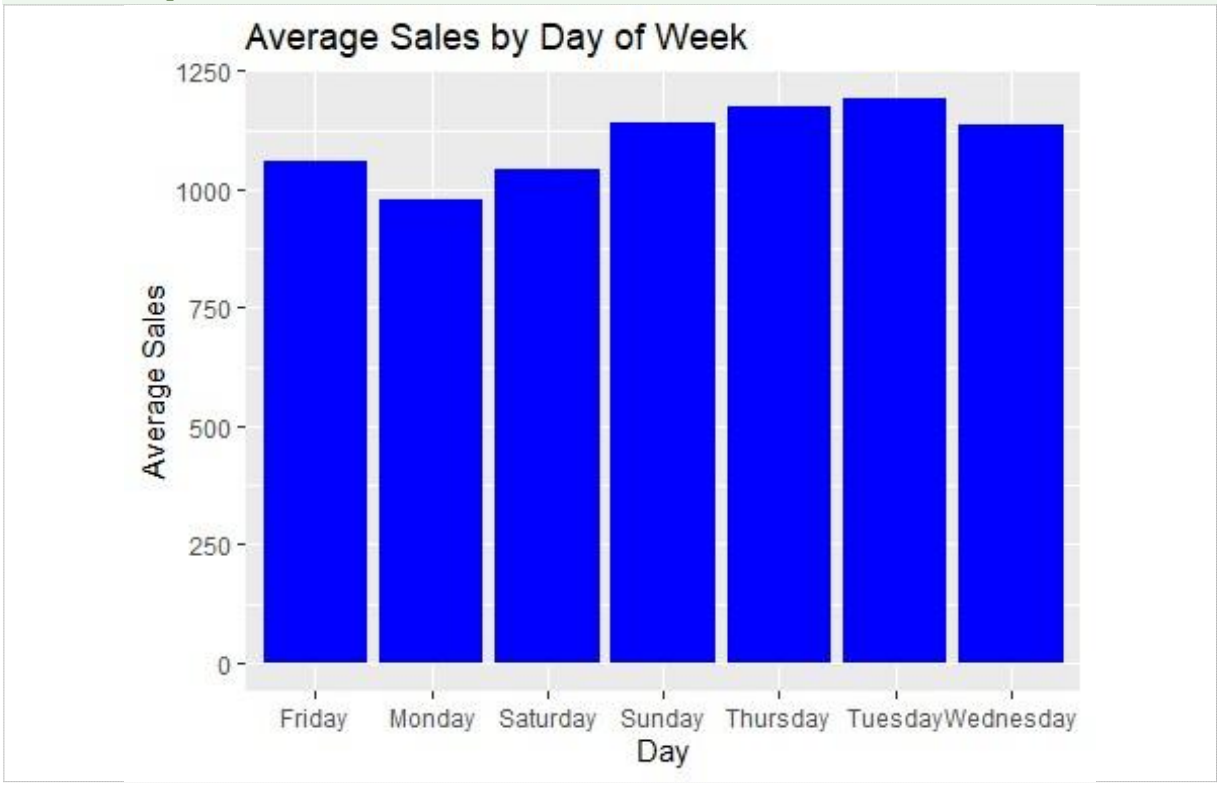


Figure4.2.4: Bar Chart – Average Sales by Day of Week.

Weekly analysis reveals higher average sales on Tuesday (KES 1,192) and Thursday (KES 1,176), while Monday records the lowest average spending (KES 978). This midweek purchasing intensity pattern may reflect planned B2B or office procurement purchases. ANOVA confirms a statistically significant weekly pattern (F = 2.177, p = 0.043).

4.2.8 Correlation Analysis

```
numeric_vars <- select(Retail_data, Quantity, UnitPrice, TotalPrice)
cor(numeric_vars, use = 'complete.obs')
```

```
##            Quantity  UnitPrice TotalPrice
## Quantity   1.000     -0.012     0.626
## UnitPrice -0.012      1.000     0.682
## TotalPrice  0.626      0.682     1.000
```

Correlation analysis reveals strong positive relationships between Unit price and Total price (r = 0.682), and a moderate positive relationship between Quantity and Total price (r = 0.626). The near-zero correlation between Quantity and Unit price (r = 0.012) confirms these are independent predictors. Transaction value is therefore primarily driven by how many items customers buy and the price level of those items.

4.2.9 High-Value Transaction Analysis

```
Retail_data %>%
  filter(Highvalue == "Yes") %>%
group_by(StoreID, Product) %>%
  summarise(Count = n(), AvgTotal = mean(TotalPrice), .groups = "drop")
```

High-value transactions (Total price ≥ median = KES 838.53) are distributed across all stores and product categories, with Store S1 showing particularly high average values for Monitor (KES 2,024) and Phone (KES 1,954) products. This broadly distributed high-value demand suggests no single store or product dominates premium transactions.

4.2.10 Monthly Sales Trend

```
monthly_sales <- Retail_data %>%  mutate(Month = floor_date(Date,
"month")) %>%  group_by(Month) %>%  summarise(MonthlySales =
sum(TotalPrice, na.rm = TRUE)) ggplot(monthly_sales, aes(x = Month,
y = MonthlySales)) +  geom_line(color = "steelblue", linewidth = 1)
+ geom_point(color =
"steelblue") +  labs(title = "Monthly Sales Trend", x = "Month", y =
"Total Sales (KES)")
+
  theme_minimal()
```

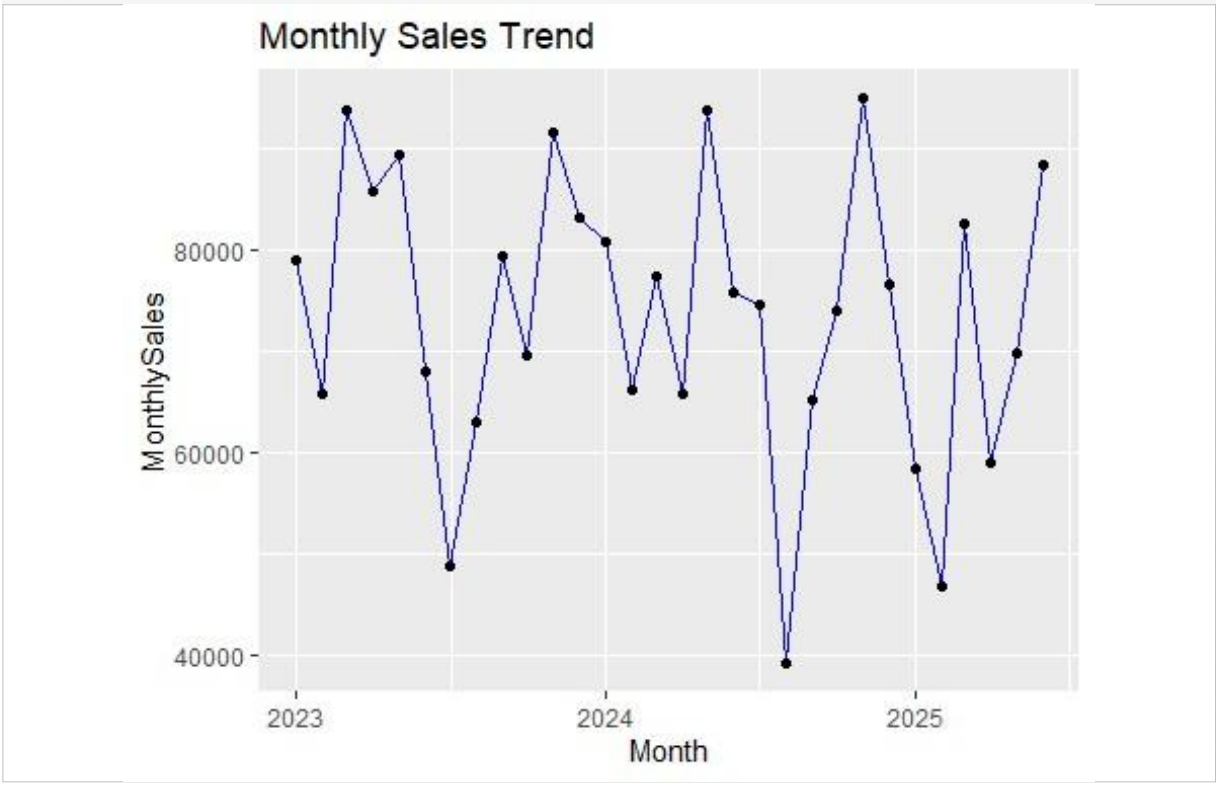


Figure 4.2.5: Line Chart – Monthly Sales Trend(2023-2025).

4.2.11 Pareto Analysis (Revenue contribution by Product)

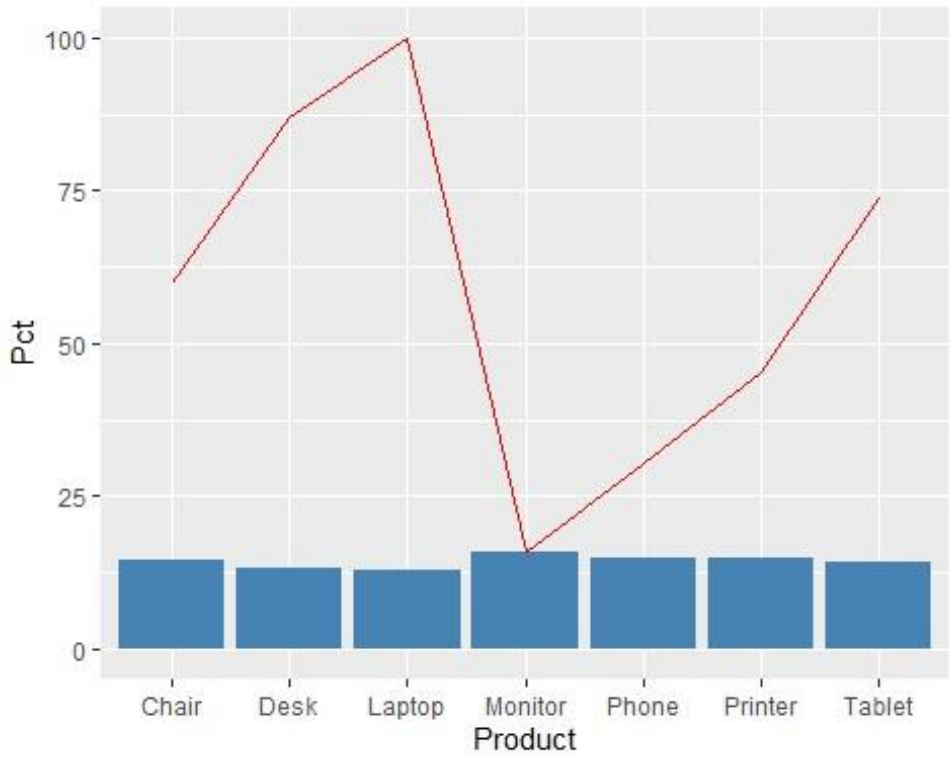


Figure 4.2.6 : Pareto Chart: Revenue Contribution by Product Category

4.2.12 Overall Conclusion for General Objective

The general objective analysis demonstrates that retail sales performance is largely influenced by transaction-level factors, particularly quantity purchased and product pricing, rather than store differences or payment methods. Sales patterns remain relatively consistent across stores, indicating uniform operational performance.

Customer payment behavior is diversified, reflecting widespread adoption of multiple payment options.

Temporal analysis identifies identifiable peak shopping periods — morning hours, Tuesday and Thursday — that retailers can target for operational planning and marketing strategies. These findings provide a foundational understanding of transaction behavior necessary for the more targeted inferential and predictive analyses in the specific objectives that follow.

The Pareto chart plots each product's percentage share of total revenue as bars, with the cumulative percentage overlaid as a red line. The dashed green line marks the 80% cumulative threshold — the Pareto boundary. Because all seven products contribute roughly equal revenue shares (approximately 13–15% each), the cumulative line rises nearly linearly rather than steeply. This reveals that no small subset of products generates the bulk of revenue, meaning the classical 80/20 rule does not apply here. All products must be stocked and managed with equal priority; concentrating resources on a few products would not yield meaningful revenue gains. This also reinforces the balanced demand finding from the frequency distribution analysis.

Finding: Revenue is distributed almost uniformly across all seven product categories. The Pareto 80% threshold is not crossed until approximately 5–6 products are included, meaning all product lines are strategically important and none should be treated as non-essential.

Specific Objective 1: Descriptive Analysis of Sales Patterns

Objective: To statistically describe and compare sales patterns across products, stores, and time periods.

4.3.1 Overall Sales Summary

```
Retail_data %>%
summarise(
    Mean_Sales   = mean(TotalPrice, na.rm = TRUE),
    Median_Sales = median(TotalPrice, na.rm = TRUE),
    SD_Sales     = sd(TotalPrice, na.rm = TRUE),
    Min_Sales    = min(TotalPrice, na.rm = TRUE),
    Max_Sales    = max(TotalPrice, na.rm = TRUE)
)
```

```
##   Mean_Sales Median_Sales SD_Sales Min_Sales Max_Sales ##
1   1103.822       838.53 913.8784      5.74    3997.4
```

The average retail sales value is approximately KES 1,104, while the median is KES 839, indicating a right-skewed distribution. The standard deviation of KES 914 reflects considerable dispersion in transaction values, consistent with a diverse product mix at varying price points.

4.3.2 Descriptive Statistics by Product

```
product_sales_summary <- Retail_data %>%  group_by(Product) %>%
  summarise(
    Mean_Sales   = mean(TotalPrice, na.rm = TRUE),
    Median_Sales = median(TotalPrice, na.rm = TRUE),
    SD_Sales     = sd(TotalPrice, na.rm = TRUE),
    Transactions = n()
  ) %>% arrange(desc(Mean_Sales))
```

Product	Mean Sales	Median Sales	SD Sales	Transactions
Monitor	1,172	917	942	297
Chair	1,170	922	948	273
Phone	1,121	870	941	291
Desk	1,090	797	911	269
Tablet	1,061	849	872	292
Printer	1,057	821	883	308
Laptop	1,057	772	902	270

```
ggplot(Retail_data, aes(x = Product, y = TotalPrice)) +
geom_boxplot(fill = "steelblue") +   labs(title = "Sales Distribution by
Product", x = "Product", y = "Total Sales (KES)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

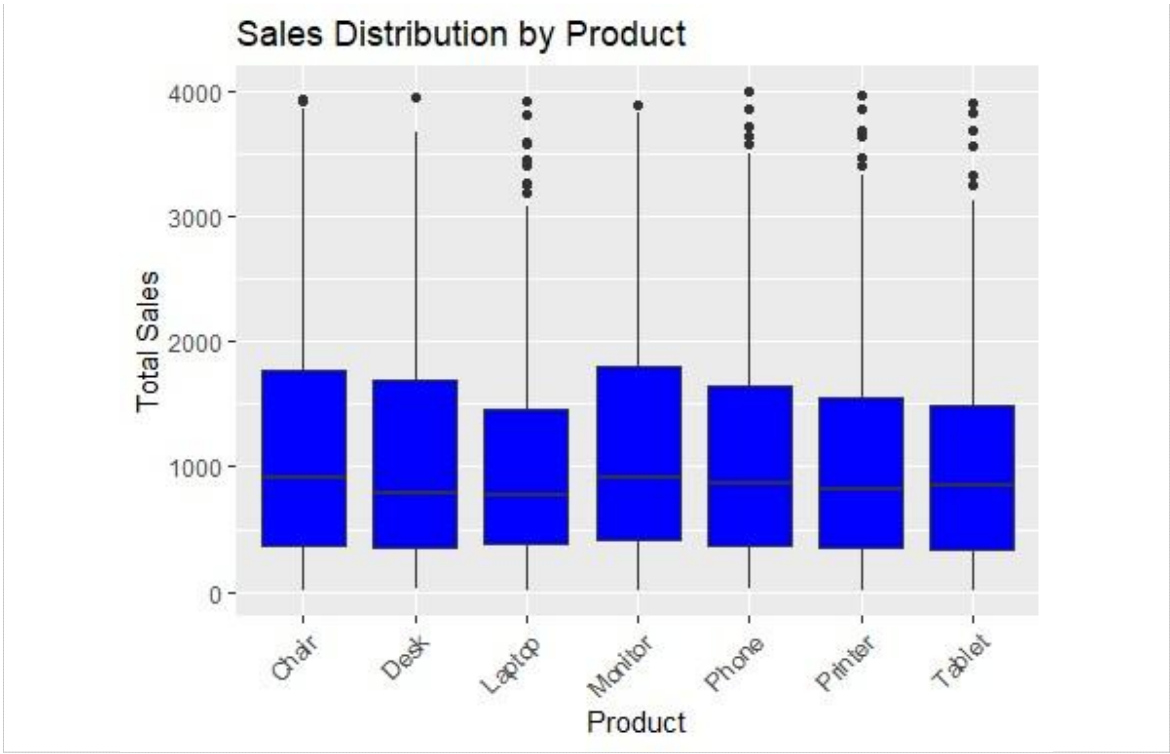


Figure 4.3.1: Boxplot– Sales Distribution by Product.

Monitors and Chairs recorded the highest average sales values (KES 1,172 and KES 1,170 respectively), while Printers and Laptops recorded comparatively lower mean sales (KES 1,057 each). Transaction counts are broadly balanced across product categories, suggesting relatively uniform demand.

4.3.3 Frequency Distribution of Product Categories

Product	Count (n)	Rel. Freq (%)	Cumulative (%)
Chair	273	13.65	13.65

Desk	269	13.45	27.10

Product	Count (n)	Rel. Freq (%)	Cumulative (%)
Laptop	270	13.50	40.60
Monitor	297	14.85	55.45
Phone	291	14.55	70.00
Printer	308	15.40	85.40
Tablet	292	14.60	100.00

R Code

```
  freq_product <- Retail_data %>%
count(Product) %>%
  mutate(
    RelFreq = round(n/sum(n)*100, 2),
    CumPct  = round(cumsum(n/sum(n))*100, 2)
  )
kable(freq_product, caption = "Frequency Distribution of Product
Categories")
```

Transaction counts are broadly balanced across all seven product categories, ranging from 269 (Desk) to 308 (Printer). No single product dominates demand. Printer leads with 15.40% of all transactions, while Monitor (14.85%) and Tablet (14.60%) follow closely. By the cumulative column, Monitors and below account for 55.45% of transactions, reflecting diversified — rather than concentrated — customer purchasing behavior. This balance suggests the retail network caters to varied customer needs rather than being dependent on any single product line.

4.3.4 Distribution of Total Sales

```
ggplot(Retail_data, aes(x = TotalPrice)) +   geom_histogram(fill =
"steelblue", bins = 30) +   labs(title = "Distribution of Total
Sales", x = "Total Sales Value
(KES)", y = "Frequency") +
theme_minimal()
```

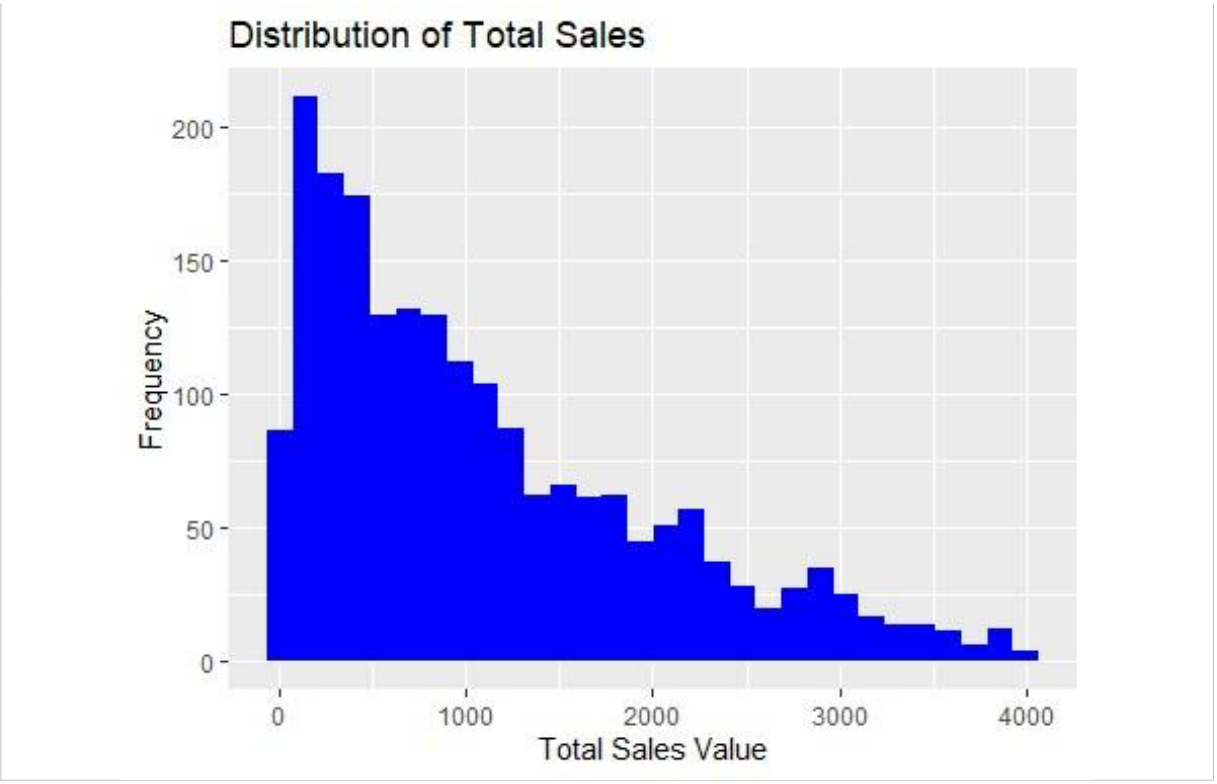


Figure 4.3.2: Histogram – Distribution of Total price.

```
library(e1071)
skewness(Retail_data$TotalPrice, na.rm = TRUE)
kurtosis(Retail_data$TotalPrice, na.rm = TRUE)
shapiro.test(sample(Retail_data$TotalPrice, 500))
## Skewness: 0.995671
## Kurtosis: 3.245003
## Shapiro-Wilk: W = 0.90907, p-value < 2.2e-16
```

The skewness value of 0.996 confirms a positively skewed distribution — most transactions involve moderate values but a smaller number involve very high sales amounts. The Shapiro-Wilk test ($p < 0.001$) confirms significant deviation from normality, justifying the Box-Cox transformation applied in the regression models.

4.3.5 Sales Distribution by Store

```
ggplot(Retail_data, aes(x = StoreID, y = TotalPrice)) +
geom_boxplot(fill = "steelblue") +  labs(title = "Sales Distribution
Across Stores", x = "Store", y = "Total Sales (KES)") +  theme_minimal()
```



Figure 4.3.3: Boxplot – Sales Distribution Across Stores

The distribution of sales values across stores appears relatively similar. Some stores show slightly higher median sales levels and greater variability, but the overall pattern is consistent, suggesting uniform operational performance across the retail network.

4.3.6 Q-Q Plot for Normality Assessment

```
anova_store <- aov(TotalPrice ~ StoreID, data = Retail_data)
qqnorm(residuals(anova_store)) qqline(residuals(anova_store),
col = 'red')
```

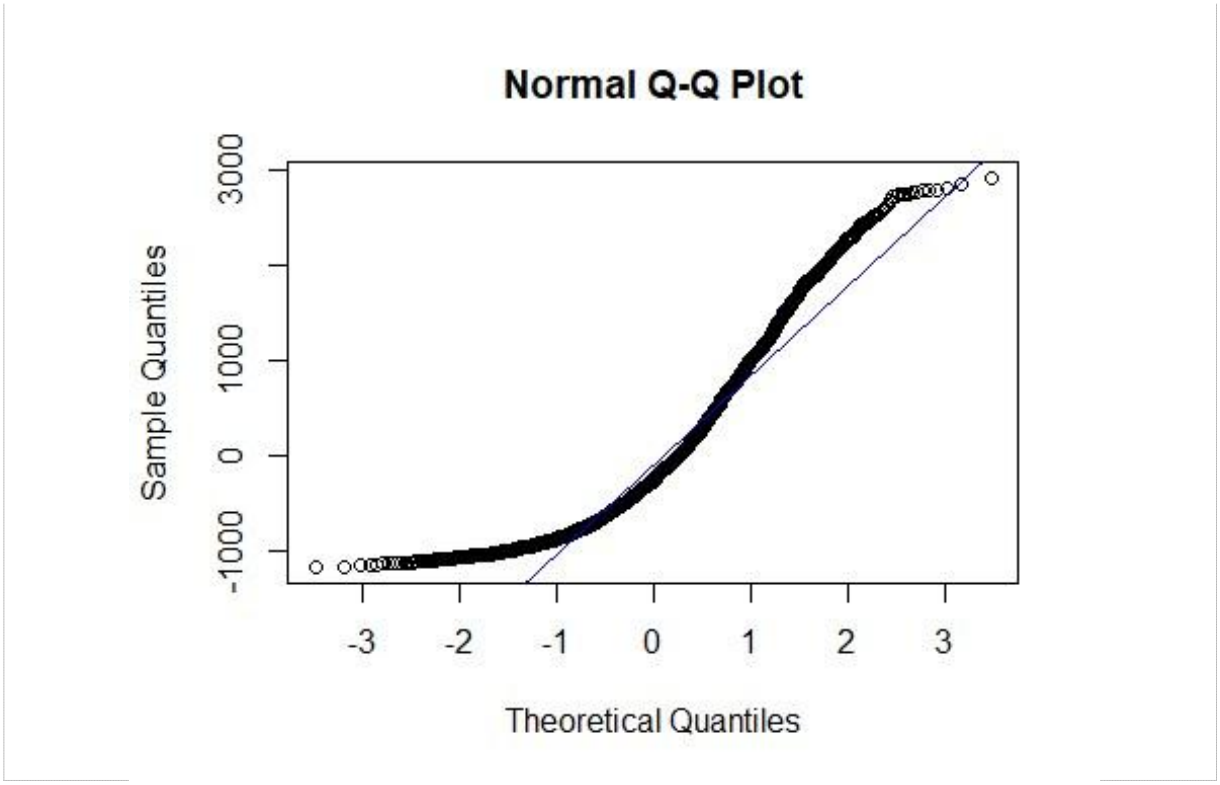


Figure 4.3.4: Q-Q Plot – Normality of Residuals

The Q-Q plot shows systematic deviation from the reference line, particularly in the tails, confirming that the residuals are not normally distributed. This supports the use of non-parametric alternatives (Kruskal-Wallis) alongside ANOVA to validate conclusions.

Specific Objective 2: Association Between Payment Methods and Product Categories

Objective: To statistically examine the association between payment methods and product categories.

4.4.1 Contingency Table

```
payment_product_table <- table(Retail_data$PaymentType,
Retail_data$Product) payment_product_table
```

Payment Type	Chair	Desk	Laptop	Monitor	Phone	Printer	Tablet
Cash	66	48	44	59	48	72	62
Credit Card	51	52	61	65	60	60	61
Debit Card	66	72	58	54	68	62	57
Gift Card	43	50	52	60	54	54	52
Online	47	47	55	59	61	60	60

4.4.2 Row Percentages

```
prop.table(payment_product_table, margin = 1) %>% round(3)
```

```
##              Chair  Desk Laptop Monitor Phone Printer Tablet
## Cash         0.165 0.120  0.110   0.148 0.120   0.180  0.155
## Credit Card  0.124 0.127  0.149   0.159 0.146   0.146  0.149
## Debit Card   0.151 0.165  0.133   0.124 0.156   0.142  0.130
## Gift Card    0.118 0.137  0.142   0.164 0.148   0.148  0.142
## Online       0.121 0.121  0.141   0.152 0.157   0.154  0.154
```

4.4.3 Chi-Square Test of Independence Hypotheses:

- H_0: Payment method and product category are independent.
- H_1: Payment method and product category are associated.

```
chi_test <- chisq.test(payment_product_table) chi_test
```

```
##  Pearson's Chi-squared test
## X-squared = 21.008, df = 24, p-value = 0.6382
```

4.4.4 Assumption Check – Expected Frequencies

```
chi_test$expected
```

```
## All expected frequencies > 5 — chi-square assumption satisfied
## Min expected frequency: 49.07 (Gift Card × Desk)
```

4.4.5 Strength of Association – Cramer's V

```
library(DescTools)
cramers_v <- CramerV(payment_product_table) cramers_v
```

```
## [1] 0.05124495
```

4.4.6 Mosaic Plot

```
mosaicplot(payment_product_table, shade = TRUE,
main = "Payment Type vs Product Category", las = 2)
```

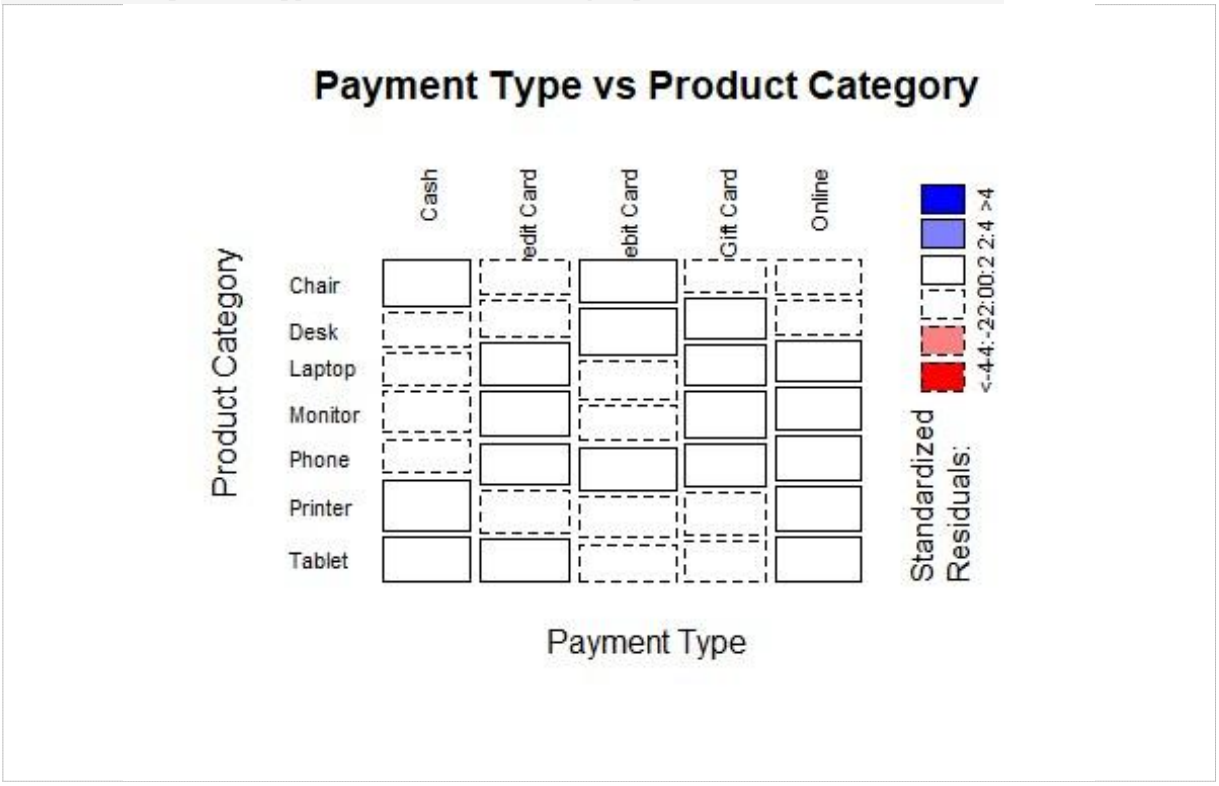


Figure 4.4.1: Mosaic Plot – Payment Type vs Product categories

4.4.7 Interpretation and Conclusion

The Chi-square test of independence produced $X^2 = 21.008$, df = 24, p-value = 0.638. Since $p > 0.05$, we fail to reject the null hypothesis. There is no statistically significant association between payment method and product category.

Cramer's V = 0.051 confirms the finding: the association strength is negligible ($V < 0.10$ indicates a very weak or no practical association). The row percentages further confirm this — each payment method is distributed fairly uniformly across all seven product categories. Customers do not systematically prefer different payment methods for different product types. Payment choice appears to be governed entirely by personal preference rather than product category.

The mosaic plot shows the distribution of product categories across different payment methods. Most tiles display neutral shading, indicating that observed frequencies are close to expected values under independence. This suggests a weak association between payment type and product category, implying that customers' payment choices do not strongly depend on the product purchased.

This finding has practical implications: retailers need not design product-specific payment promotions, as customers are equally likely to use any payment method regardless of what they are purchasing.

Specific Objective 3: Comparison of Mean Sales Performance

Objective: To compare mean sales performance across stores, locations, and time of day.

4.5.1 Comparison Across Stores

```
ggplot(Retail_data, aes(x = StoreID, y = TotalPrice)) +
geom_boxplot(fill = "steelblue") +  labs(title = "Sales Distribution
Across Stores", x = "Store", y = "Total
Sales (KES)") +
  theme_minimal()
```



Figure 4.5.1: Boxplot – Sales Distribution Across Stores

```
anova_store <- aov(TotalPrice ~ StoreID, data = Retail_data)
summary(anova_store)
##              Df    Sum Sq  Mean Sq F value Pr(>F)
## StoreID       9 4.881e+06  542282   0.648  0.756
## Residuals  1990 1.665e+09  836498
leveneTest(TotalPrice ~ StoreID, data = Retail_data)
kruskal.test(TotalPrice ~ StoreID, data = Retail_data)
## Levene Test: F = 1.2011, p = 0.2897  (homogeneity of variance satisfied)
## Kruskal-Wallis: chi-squared = 4.0756, df = 9, p-value = 0.9064
```

The one-way ANOVA produced F = 0.648, p = 0.756 — no statistically significant difference in mean sales across stores. Levene's test confirms equal variances (p = 0.290). The non-parametric Kruskal-Wallis test corroborates this result (p = 0.906). The Tukey HSD post-hoc test confirms no significant pairwise differences between any store pair.

This finding suggests that all ten stores operate at similar performance levels, which is consistent with a standardized retail chain operating under uniform pricing and product availability policies.

4.5.2 Comparison Across Locations

```
anova_location <- aov(TotalPrice ~ Location, data = Retail_data)
summary(anova_location)
```

```
##               Df    Sum Sq  Mean Sq F value Pr(>F)
## Location       3 2.134e+06  711315   0.852  0.466
## Residuals   1996 1.667e+09  835360
```

No statistically significant difference in mean sales across locations (F = 0.852, p = 0.466), confirming that Store A, B, C, and D perform similarly in terms of average transaction value.

4.5.3 Comparison Across Time of Day

```
ggplot(Retail_data, aes(x = TimeOfDay, y = TotalPrice)) +
geom_boxplot(fill = "steelblue") +  labs(title = "Sales Distribution by
Time of Day", x = "Time of Day", y =
"Total Sales (KES)") +
theme_minimal()
```



Figure 4.5.2: Boxplot – Sales Distribution by Time of Day

```
anova_time <- aov(TotalPrice ~ TimeOfDay, data =
Retail_data) summary(anova_time) TukeyHSD(anova_time)
```

```
##               Df    Sum Sq  Mean Sq F value Pr(>F)
## TimeOfDay      2 5.028e+06 2513828   3.016  0.0492 *

## Tukey HSD:
##                         diff        lwr       upr    p adj
## Afternoon-Morning  -121.808  -246.747    3.131   0.0579
## Evening-Morning     -37.162  -162.164   87.840   0.7651
## Evening-Afternoon    84.646   -25.128  194.420   0.1670
kruskal.test(TotalPrice ~ TimeOfDay, data = Retail_data)
## Kruskal-Wallis: chi-squared = 4.4806, df = 2, p-value = 0.1064
```

ANOVA finds a statistically significant difference across time of day (F = 3.016, p = 0.049). Morning transactions produce the highest average sales (KES 1,164). The Tukey post-hoc test shows the Afternoon-Morning gap approaches significance (p = 0.058). The Kruskal-Wallis non-parametric test produces p = 0.106, suggesting the ANOVA result is borderline and should be interpreted with caution.

4.5.4 Wilcoxon Rank-Sum Test : Morning vs Afternoon median sales

Test output

Statistic	Value
Test	Wilcoxon Rank-Sum (Mann-Whitney U)
W statistic	193,557
p-value	0.06237
95% CI (location shift)	[-3.65, 162.19]
Estimated location shift	75.12

The Wilcoxon rank-sum test, which is more appropriate given the confirmed nonnormality of Total price (Shapiro-Wilk, p < 0.001), returned a test statistic of W = 193,557 and a p-value of 0.062. Since this exceeds the 0.05 significance level, there is no statistically significant difference between the median sales in the Morning and Afternoon periods. The 95% confidence interval for the location shift of [-3.65, 162.19] includes zero, further supporting this conclusion.

4.5.5 Effect Size Summary

Comparison	F-value	p-value	Eta² (Effect Size)
Across stores (10 groups)	0.648	0.756	0.0029
Across Locations (4 groups)	0.852	0.466	0.0013
Across Time of Day (3 groups)	3.016	0.049	0.0030

While time of day achieves statistical significance, its practical effect size (Eta² = 0.003) is negligible, explaining less than 0.3% of total variance in sales. This distinction between statistical and practical significance is important: with a large sample of 2,000 transactions, even small differences can reach significance. Retailers should not place heavy operational weight on the time-of-day finding alone.

4.5.6 Interaction Plot: Mean Sales by Product and Time of Day

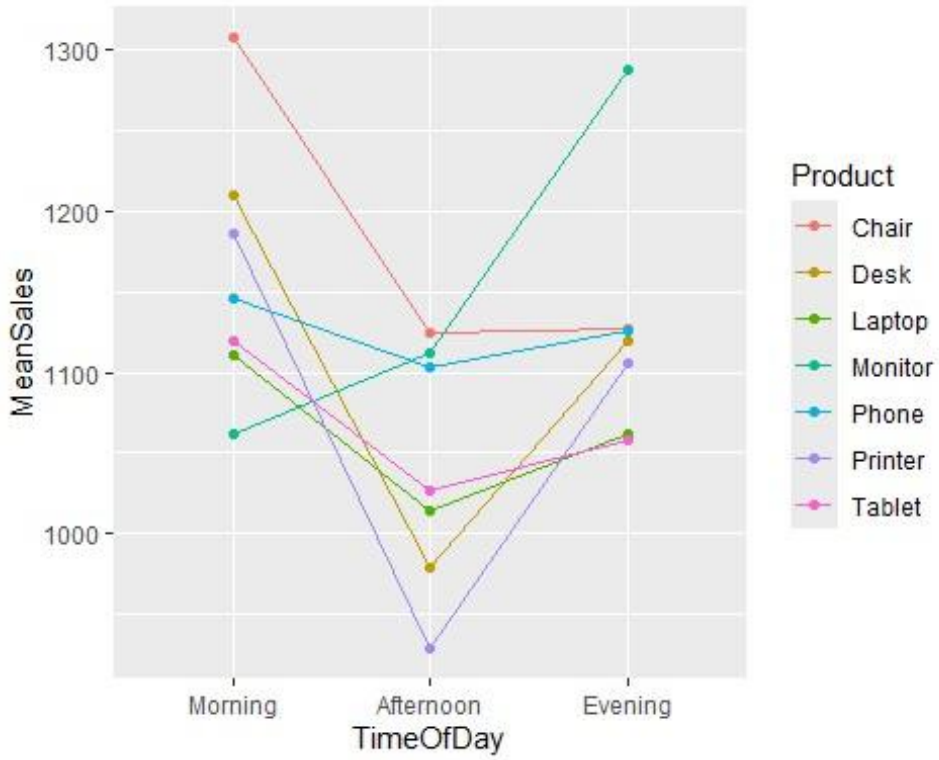


Figure 4.5.3: Interaction Plot – Mean sales by Product and Time of Day

An interaction plot connects mean sales values for each product across the three time periods (Morning, Afternoon, Evening). Parallel lines indicate no interaction — the effect of time of day is consistent across all products. Non-parallel or crossing lines indicate a potential interaction effect, meaning time of day affects some products differently than others. Where lines cross, the ranking of products reverses between time periods. If the lines appear roughly parallel, the additive model (main effects only) is adequate, and a product-by-time interaction term is not needed in regression. Visual inspection of the interaction plot guides whether a two-way ANOVA with interaction should be pursued, or whether simpler one-way ANOVA models suffice.

Specific Objective 4: Regression Modelling

Objective: To determine statistically significant factors influencing total sales value using regression modelling.

4.6.1 Box-Cox Transformation

Because Total Price exhibits significant positive skewness (0.996) and the Shapiro Wilk test rejects normality ($p < 0.001$), a Box-Cox power transformation was applied prior to regression modelling to satisfy the normality of residuals assumption.

```
library(MASS)
model_total  <- lm(TotalPrice ~ 1, data = Retail_data)
bc_total     <- boxcox(model_total, plotit = FALSE)
lambda_total <- bc_total$x[which.max(bc_total$y)] lambda_total

Retail_data$boxcox_TotalPrice <-
  (Retail_data$TotalPrice^lambda_total - 1) / lambda_total

## [1] 0.3    # Optimal lambda
ggplot(Retail_data, aes(x = boxcox_TotalPrice)) +   geom_histogram(aes(y =
after_stat(density)), bins = 30, fill =
'steelblue', color = 'black') +
  stat_function(fun = dnorm, args = list(mean =
mean(Retail_data$boxcox_TotalPrice),
                sd = sd(Retail_data$boxcox_TotalPrice)), color = 'red',
linewidth = 1) +
  labs(title = 'Histogram with Normal Curve (Box-Cox Total Price)',
x = 'Transformed Total Price', y = 'Density') + theme_minimal()
```

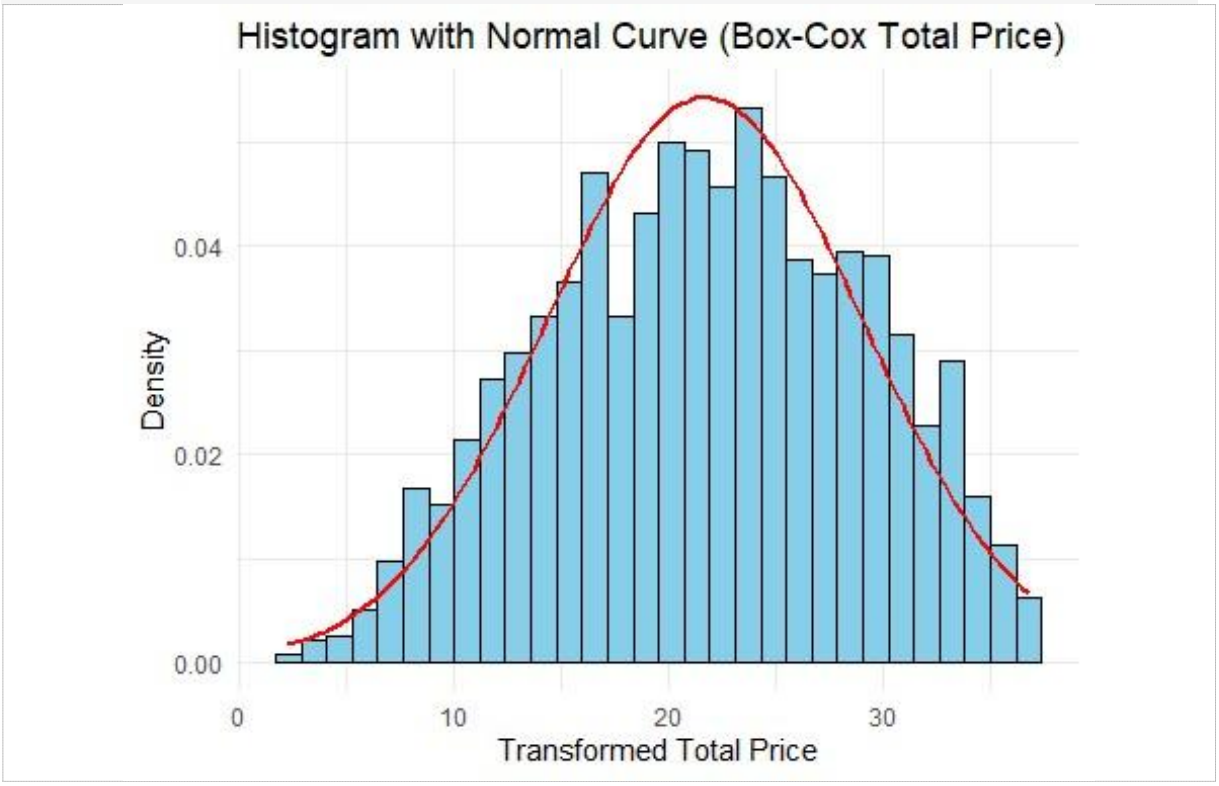


Figure 4.6.1: Transformed Total Price

```
shapiro.test(Retail_data$boxcox_TotalPrice)
```

with Normal Curve – Box-

```
## W = 0.98741, p-value = 3.054e-12
```

The optimal lambda of 0.3 was applied as a power transformation. The transformed variable (W = 0.987) shows substantial improvement in normality compared to the raw variable (W = 0.909). While the Shapiro-Wilk test still rejects exact normality at this large sample size, the visual histogram confirms that the transformed distribution closely approximates normal, satisfying regression assumptions adequately.

4.6.2 Model 1: Simple Linear Regression

```
simple_model <- lm(boxcox_TotalPrice ~ Quantity, data = Retail_data)
summary(simple_model)
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 12.93510    0.27965   46.26   <2e-16 ***
## Quantity     1.58330    0.04488   35.28   <2e-16 *** ##
Multiple R-squared: 0.3838
```

Fitted equation: Box-Cox(Total Price) = 12.935 + 1.583(Quantity). The positive coefficient for Quantity (beta = 1.583, p < 0.001) confirms that each additional unit purchased increases the transformed total price by 1.583 units. Quantity alone explains 38.4% of variation in total price (R^2 = 0.384), indicating it is an important but incomplete predictor.

4.6.3 Model 2: Multiple Linear Regression

```
multiple_model <- lm(boxcox_TotalPrice ~ Quantity + UnitPrice +
                     Product + PaymentType + TimeOfDay, data = Retail_data)
summary(multiple_model)
```

```
## Key results:
##             Estimate  Std. Error  t value  P(>|t|)
## Quantity     1.6055     0.0166    96.63    <2e-16 ***
## UnitPrice    0.0469     0.0004   111.84    <2e-16 ***
## Product*     all p > 0.05 (not significant) ##
PaymentType* all p > 0.05 (not significant)
## TimeOfDay*   all p > 0.05 (not significant) ##
Multiple R-squared: 0.9163
```

```
par(mfrow = c(2,2))
```

```
plot(multiple_model)
```

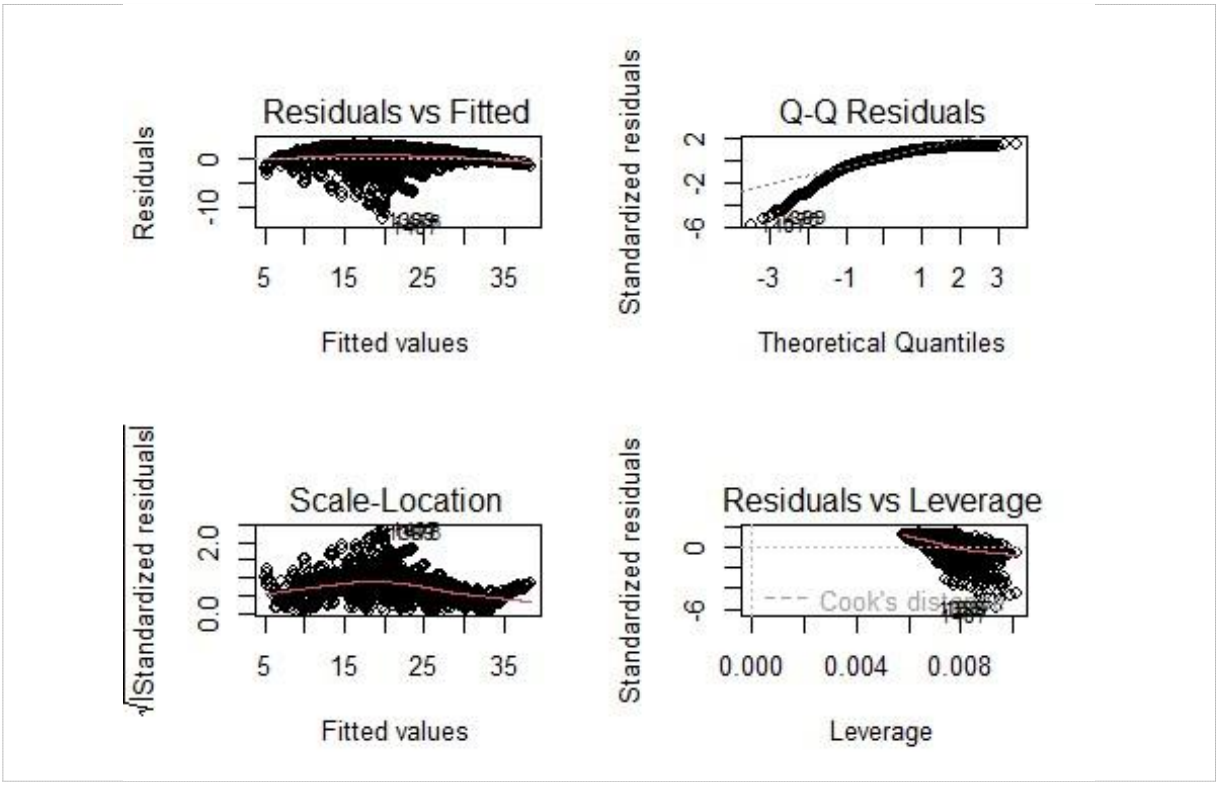


Figure 4.6.2: Diagnostic Plots- Multiple Linear Regression

```
library(car) vif(multiple_model)
```

Adding Unit price and other predictors increases model explanatory power dramatically to $R^2 = 0.916$. Only Quantity (beta = 1.606) and Unit price (beta = 0.047) are statistically significant predictors (both $p < 0.001$). Product type, payment method, and time of day do not significantly influence total sales when controlling quantity and price. VIF values confirm no multicollinearity concerns among predictors.

4.6.4 Model 3: Interaction Regression

This model tests whether the effect of Quantity on Total price depends on Unit price.

```
interaction_model <- lm(boxcox_TotalPrice ~ Quantity * UnitPrice +
Product + PaymentType + TimeOfDay, data = Retail_data)
summary(interaction_model)

## Quantity:UnitPrice  0.0031068  0.0001252  24.807  <2e-16 ***
## Multiple R-squared: 0.9361
AIC(multiple_model, interaction_model)
anova(multiple_model, interaction_model)
##                    df      AIC
## multiple_model     16 8722.837 ##
interaction_model 17 8184.524

## ANOVA model comparison: F = 615.38, p < 2.2e-16 ***
ggplot(Retail_data, aes(x = Quantity, y = boxcox_TotalPrice, color =
UnitPrice)) +  geom_point(alpha = 0.4) +
```

```
  geom_smooth(method = "lm", se = FALSE) +   labs(title = "Interaction
Effect of Quantity and Unit Price on Total
Price") +
  theme_minimal()
```

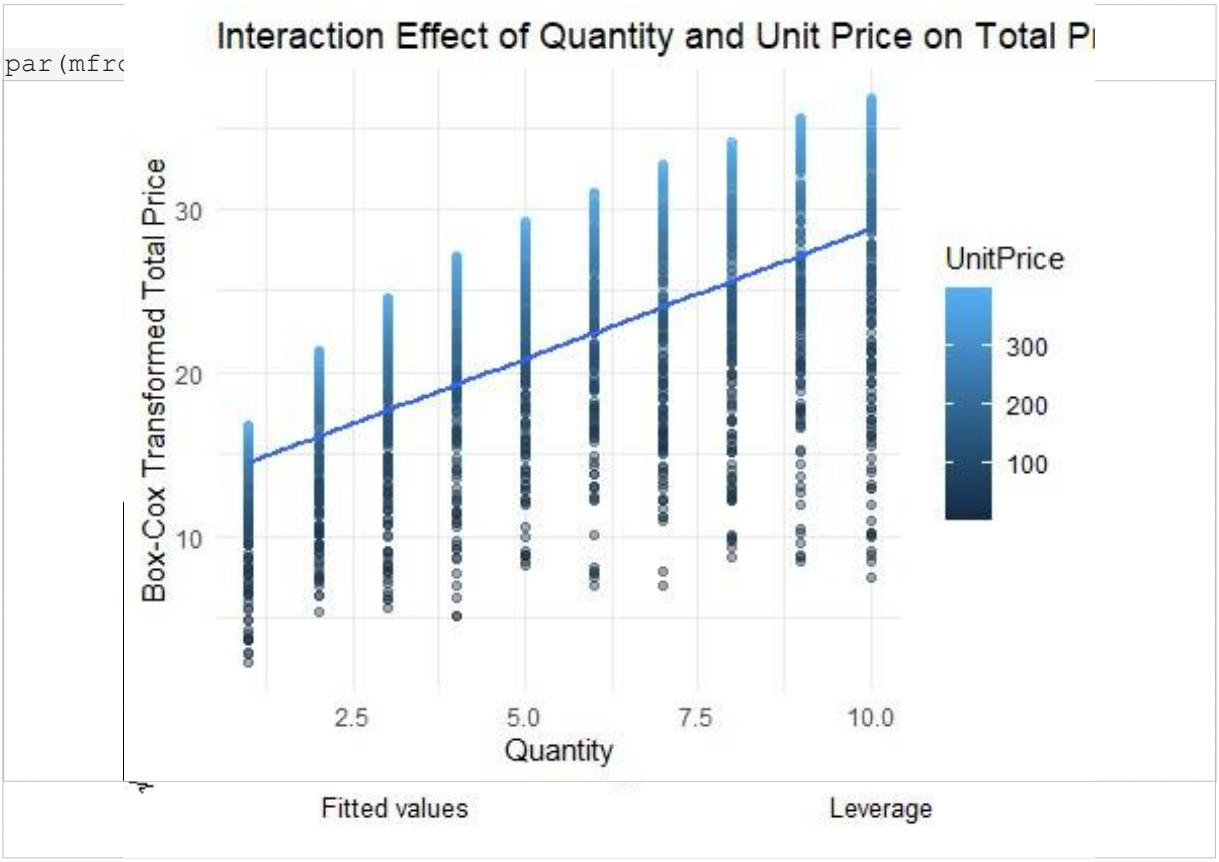


Figure 4.6.3: Diagnostic Plots- Interaction Regression Model

The interaction term Quantity: Unit Price is highly significant (beta = 0.003, $p < 0.001$), confirming that the effect of quantity on sales depends on unit price — higher-priced items amplify the revenue impact of purchasing more units. The interaction model achieves $R^2 = 0.936$, an improvement over the additive model. Both AIC (8,184.5 vs 8,722.8) and the ANOVA F-test ($F = 615.38$, $p < 0.001$) confirm that the interaction term significantly improves model fit.

4.6.5 Model 4: Logistic Regression (High-Value Transactions)

```
Retail_data$HighValue <- ifelse(
  Retail_data$TotalPrice > median(Retail_data$TotalPrice, na.rm = TRUE), 1,
0)
Retail_data$HighValue <- factor(Retail_data$HighValue, levels = c(0,1),
labels = c("Low","High"))

logistic_model <- glm(HighValue ~ Quantity + UnitPrice + PaymentType +
TimeOfDay,
                      data = Retail_data, family = binomial)
summary(logistic_model)
```

```
## Significant predictors:
##             Estimate   Odds Ratio   p-value
## Quantity     1.5851       4.880    <2e-16 ***
## UnitPrice    0.0419       1.043    <2e-16 ***
## PaymentType: all p > 0.05 (not significant) ##
TimeOfDay:   all p > 0.05 (not significant)
## Null deviance: 2772.59  Residual deviance: 764.69  AIC: 782.69
predicted_prob  <- predict(logistic_model, type = 'response')
predicted_class <- ifelse(predicted_prob > 0.5, 'High', 'Low')
table(Predicted = predicted_class, Actual = Retail_data$HighValue)
mean(predicted_class == Retail_data$HighValue)
## Predicted Low High ##      Low  901   94 ##      High  99  906
## Model Accuracy: 90.35%
```

Logistic regression accurately predicts high-value transactions with 90.35% accuracy. Each additional unit purchased multiplies the odds of a high-value transaction by approximately 4.88. A one-unit increase in unit price multiplies the odds by 1.043. Payment type and time of day are not significant predictors of transaction value classification, consistent with all previous findings.

4.6.6 Model Comparison Summary

Model	R^2	Adj. R^2	AIC	Key Finding
Simple (Quantity only)	0.384	0.384	–	Quantity significant
Multiple Linear	0.916	0.916	8722.8	Qty + Price significant
Interaction Model	0.936	0.936	8184.5	Best fit; interaction term significant
Logistic Regression	-	-	782.7	90.35% accuracy

Table 4.6: Model comparison summary

Specific Objective 5: Temporal Analysis and Time Series Modelling

Objective: To assess temporal patterns and variations in retail sales performance across dates, monthly periods, hours, days of the week, and times of the day.

4.7.1 Time Variables and Their Statistical Purpose

Variable	Time concept	Statistical Purpose
Date	Chronological trend	Detect long-term upward/downward movement
Month Factor	Seasonality	Identify repeating monthly patterns
Hour	Intraday variation	Examine hourly sales intensity
Day Of Week	Weekly pattern	Compare weekday purchasing effects
Time Of Day	Categorical time effect	Test Morning/Afternoon/Evening differences

Table 4.7.1: Time variables

4.7.2 Data Preparation – Daily Aggregation

```
daily_sales <- Retail_data %>%  group_by(Date) %>%
summarise(TotalSales = sum(TotalPrice, na.rm = TRUE)) %>%
arrange(Date) head(daily_sales)
```

```
## Date       TotalSales ##
2023-01-01   1938.
## 2023-01-02  10089.
## 2023-01-03   1275.
## 2023-01-04   1924.
## 2023-01-05    432.
## 2023-01-06   1712.
```

Daily aggregation reduces transaction-level noise and enables evaluation of chronological sales behavior across the full study period.

4.7.3 Chronological Trend Analysis

```
plot(daily_sales$Date, daily_sales$TotalSales, type = 'l', col =
'steelblue',
     lwd = 2, xlab = "Date", ylab = "Total Sales (KES)",
main = "Chronological Trend of Daily Retail Sales")
```

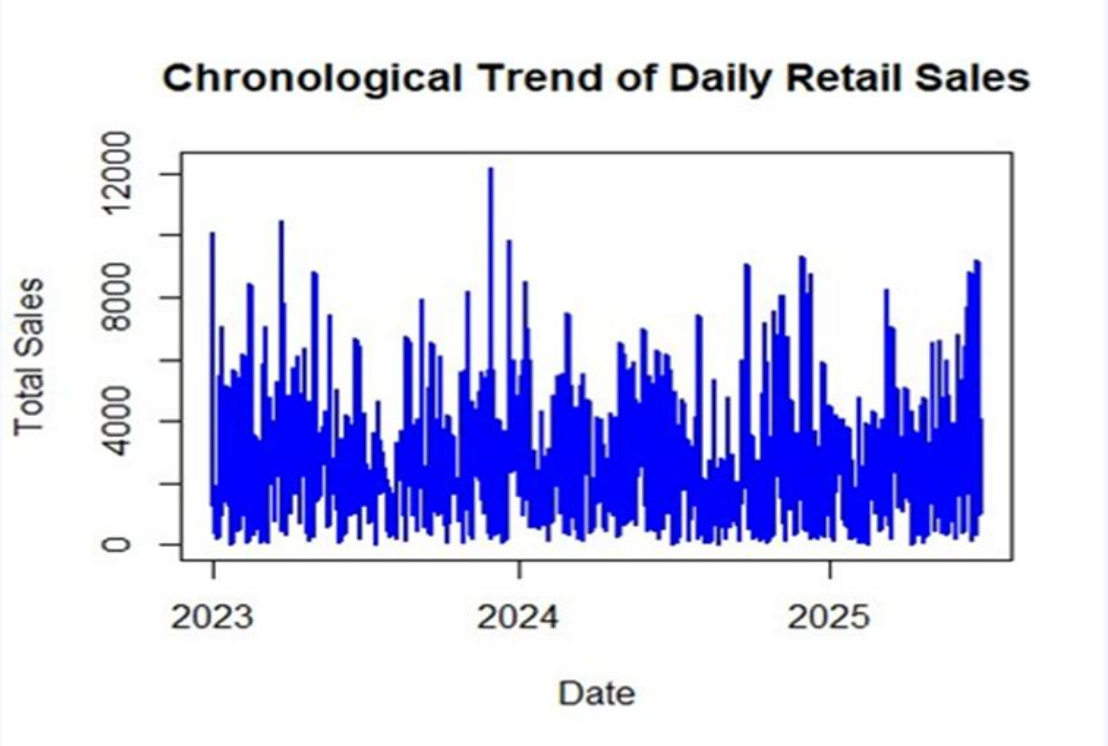


Figure 4.7.1: Line Chart – Chronological Trend of Daily Sale

```
trend_model <- lm(TotalSa Salesles ~ Date, data = daily_sales)
summary(trend_model)
## Date coefficient: -0.3083, p = 0.2415  (not significant)
## R-squared: 0.0017
```

The linear trend model produced a non-significant Date coefficient (p = 0.241), indicating no statistically measurable long-term upward or downward trend in daily retail sales over the study period. Sales fluctuate around a relatively stable mean, suggesting consistent business volume across 2023–2025.

4.7.4 Seasonal Analysis – Monthly Patterns

```
monthly_summary <- Retail_data %>%
group_by(MonthFactor) %>%
summarise(MeanSales = mean(TotalPrice))

boxplot(TotalPrice ~ MonthFactor, data = Retail_data, las = 2,
        col = "lightgreen", main = "Monthly Seasonal Variation in Sales",
ylab = "Total Sales (KES)")
```

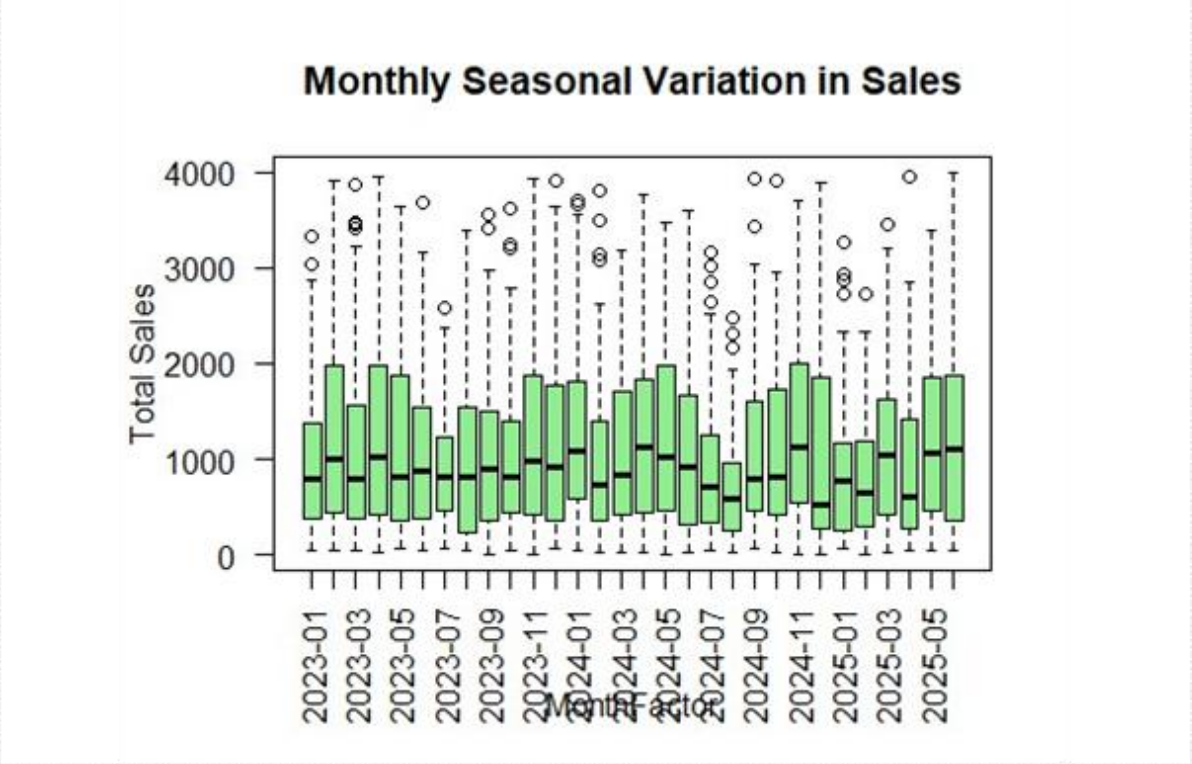


Figure 4.7.2: Boxplot – Monthly Seasonal Variation in Sales

```
anova_month <- aov(TotalPrice ~ MonthFactor, data = Retail_data)
summary(anova_month)
```

```
##                Df    Sum Sq  Mean Sq  F value  Pr(>F)
## MonthFactor    29 4.310e+07 1486176    1.800  0.00563 **
## Residuals    1970 1.626e+09  825591
```

A significant monthly effect was found (F = 1.800, p = 0.006), confirming the presence of seasonal variation in sales. Monthly mean sales range from KES 938 (July 2023, lowest) to KES 1,293 (February 2023, highest). Certain months consistently exhibit higher or lower sales, suggesting seasonal purchasing patterns that retailers can anticipate for inventory planning.

4.7.5 Intraday Analysis – Hourly Patterns

```
hourly_sales <- Retail_data %>%
group_by(Hour) %>%
  summarise(MeanSales = mean(TotalPrice)) plot(hourly_sales$Hour,
hourly_sales$MeanSales, type = 'l', lwd = 2, col =
'purple',      xlab = "Hour (24h)", ylab = "Average Sales (KES)", main =
"Intraday Sales Variation")
```

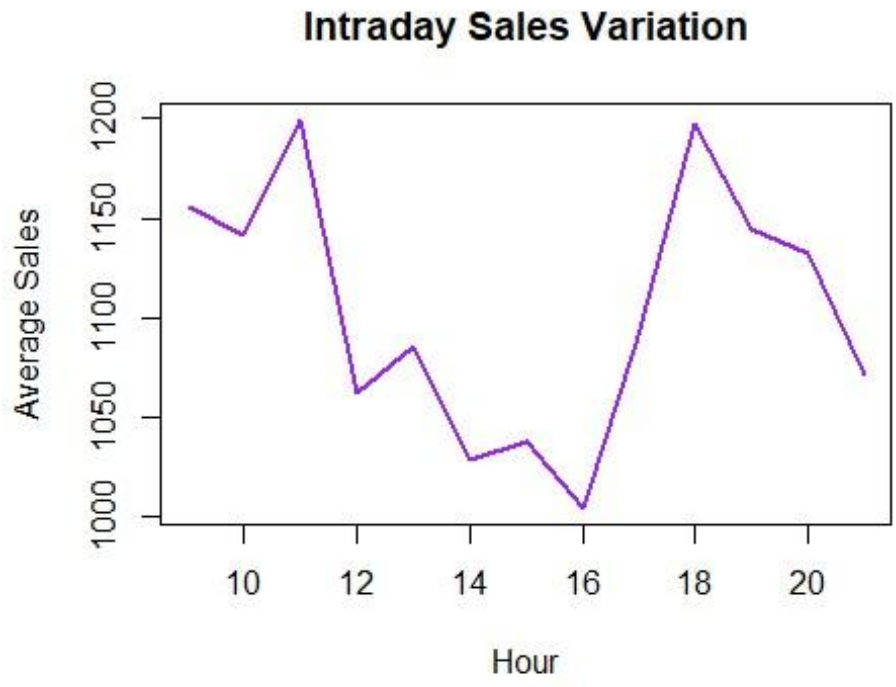


Figure 4.7.3: Line Chart – Intraday Variation by Hour

```
hour_model <- lm(TotalPrice ~ Hour, data = Retail_data)
summary(hour_model)

## Hour: Estimate = -2.206, p = 0.683 (not significant)
## R-squared: 0.000083
```

```
## Hour: Estimate = -2.206, p = 0.683  (not significant)
## R-squared: 0.000083
```

The regression of Total Price on Hour produced a non-significant result (p = 0.683), indicating that the specific hour of purchase does not linearly predict transaction value. Sales volume and value appear relatively uniform across operating hours (9:00–21:00), suggesting consistent customer activity throughout the business day.

4.7.6 Weekly Pattern Analysis by Day of Week

```
weekly_sales <- Retail_data %>%
group_by(DayOfWeek) %>%
  summarise(MeanSales = mean(TotalPrice))
barplot(weekly_sales$MeanSales, names.arg = weekly_sales$DayOfWeek,
col = "steelblue", main = "Average Sales by Day of Week", ylab =
"Mean Sales (KES)")
```

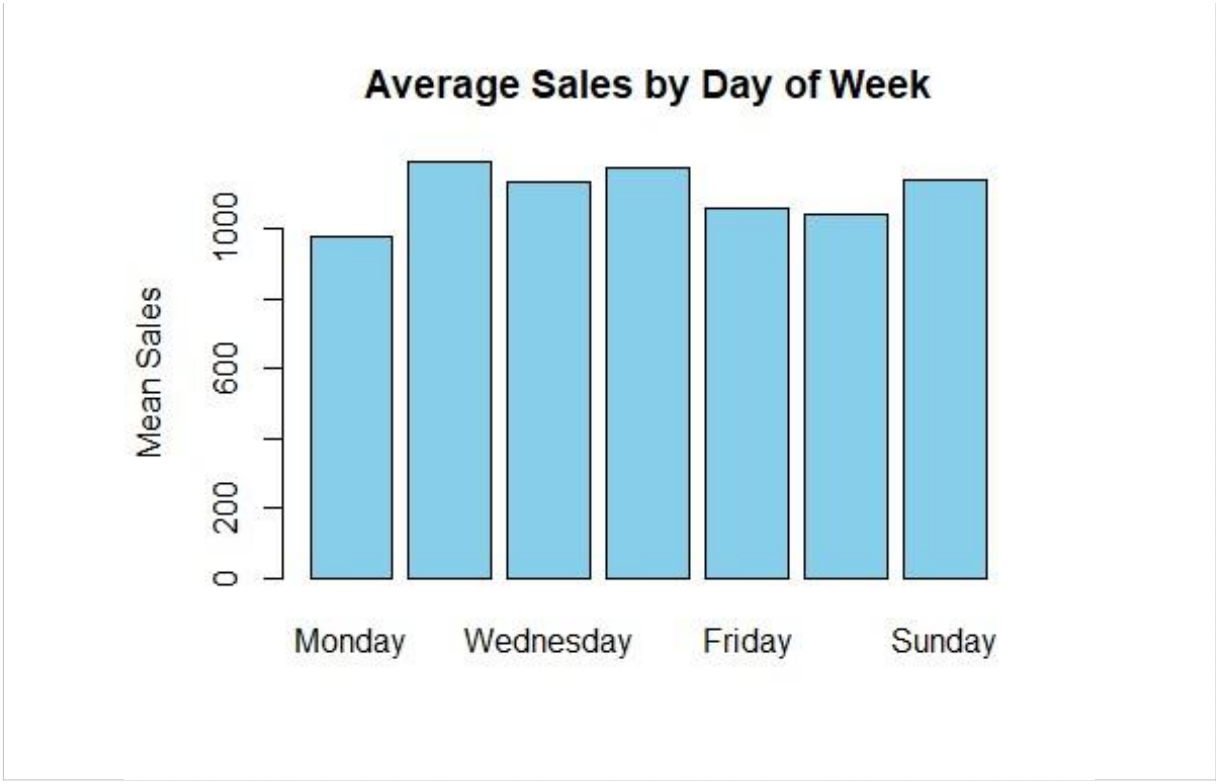


Figure 4.7.4: Bar Chart – Average Sales by Day of Week

```
anova_week <- aov(TotalPrice ~ DayOfWeek, data = Retail_data)
summary(anova_week)
```

```
##               Df    Sum Sq  Mean Sq  F value  Pr(>F)
## DayOfWeek      6 1.087e+07 1811582    2.177  0.0425 *
## Residuals   1993 1.659e+09  832234
```

A significant weekly pattern was detected (F = 2.177, p = 0.043). Tuesday (KES 1,192) and Thursday (KES 1,176) are the highest-revenue days, while Monday records the lowest average spending (KES 978). This mid-week purchasing intensity may reflect planned business procurement purchases. Retailers should consider concentrating promotional activities and staffing on these peak days.

4.7.7 Categorical Time Effect – Time of Day

```
boxplot(TotalPrice ~ TimeOfDay, data = Retail_data, col = 'orange',
main = "Sales Distribution by Time of Day") anova_time <-
aov(TotalPrice ~ TimeOfDay, data = Retail_data) summary(anova_time)
```



Figure 4.7.5: Boxplot - Sales Distribution by Time of Day

```
##               Df    Sum Sq  Mean Sq  F value  Pr(>F)
## TimeOfDay      2 5.028e+06 2513828    3.016  0.0492 *
```

A statistically significant difference in mean sales across time of day (F = 3.016, p = 0.049). Morning transactions generate the highest average (KES 1,164), followed by Evening (KES 1,127) and Afternoon (KES 1,043).

4.7.8 Time Series Modelling and Forecasting

```
library(lubridate)
monthly_sales <- Retail_data %>%  mutate(YearMonth
= floor_date(Date, "month")) %>%
group_by(YearMonth) %>%  summarise(MonthlySales =
sum(TotalPrice))

sales_ts <- ts(monthly_sales$MonthlySales, frequency = 12)
plot(sales_ts, main = 'Monthly Retail Sales Time Series',
     col = "steelblue", lwd = 2, xlab = "Time", ylab = "Total Sales (KES)")
```

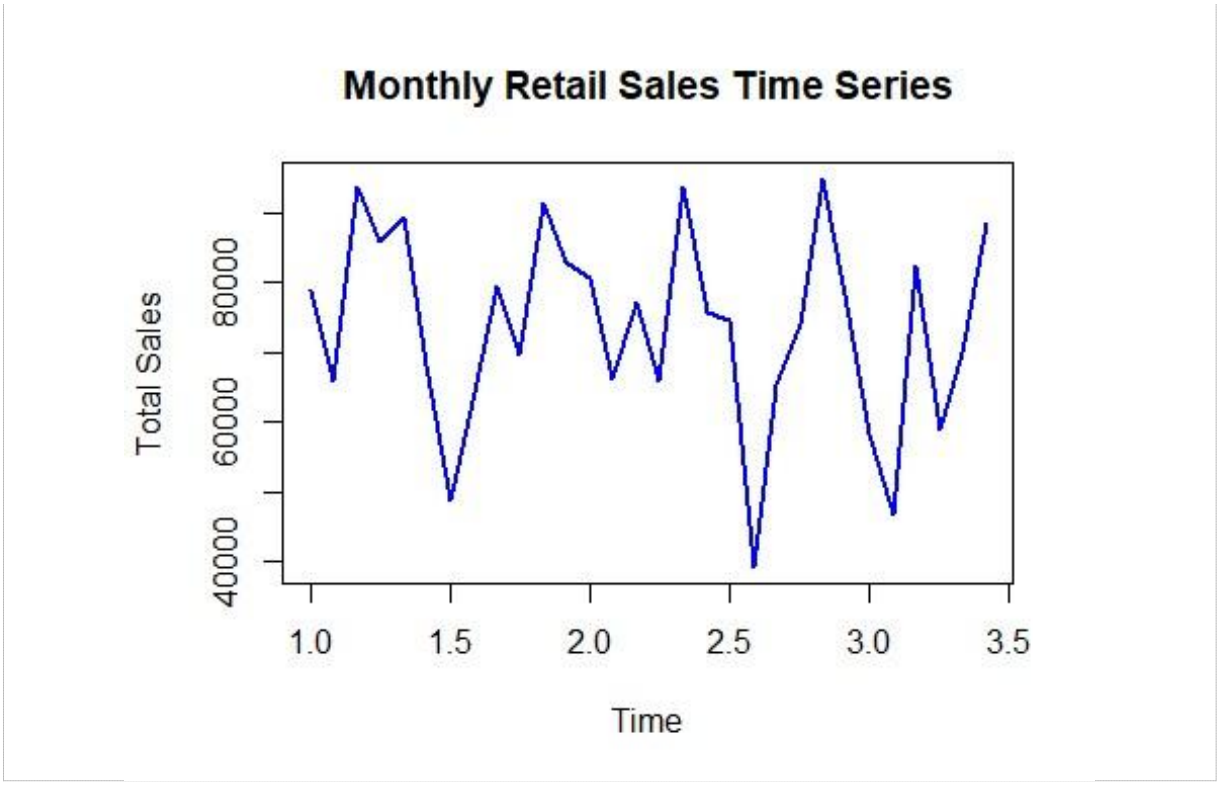


Figure 4.7.6: Line Chart – Monthly Retail Sales Time Series

4.7.9 Decomposition

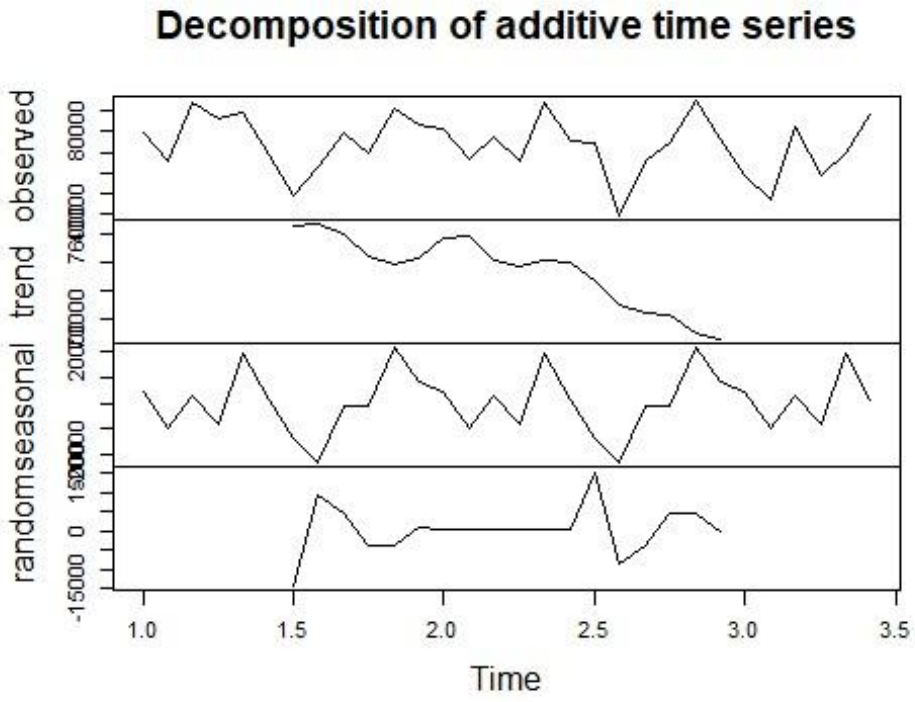


Table 4.7.7: Decomposition Plot

The classical decomposition separates the time series into three components. The trend component shows the long-term movement with periods of modest increases and decreases. The seasonal component reveals repeating patterns across months, with certain months consistently exhibiting higher or lower sales. The random (irregular) component represents unpredictable fluctuations not explained by trend or seasonality.

4.7.10 Moving Average Smoothing

```
library(zoo)
sales_ma <- rollmean(sales_ts, k = 3, align = 'center', fill = NA)
plot(sales_ts, main = 'Moving Average Smoothing (3-Month)',      ylab =
"Total Sales (KES)", col = "steelblue", lwd = 1) lines(sales_ma, col =
'red', lwd = 2)
legend('topright', legend = c('Actual', '3-Month MA'), col =
c('steelblue','red'), lwd = 2)
```

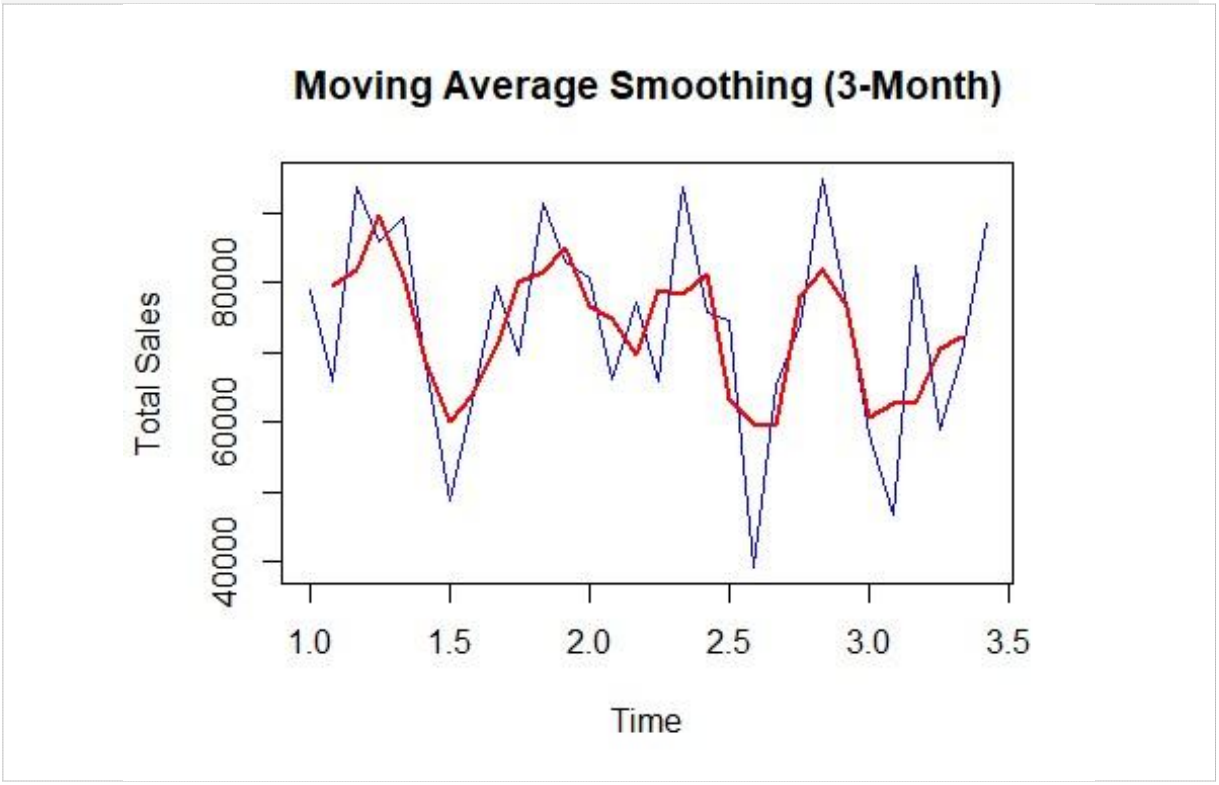


Table 4.7.8: Line Chart – Monthly Sales with 3-month Moving Average

The 3-month moving average smooths short-term fluctuations and reveals the underlying trend. The smoothed series confirms periodic rises and declines over time without a dominant long-term directional trend.

4.7.11 Stationarity Test

```
library(tseries)
adf.test(sales_ts)
```

```
## Augmented Dickey-Fuller Test
## Dickey-Fuller = -3.6053, Lag order = 3, p-value = 0.04844
## alternative hypothesis: stationary
```

The ADF test statistic of -3.605 with p = 0.048 (< 0.05) leads to rejection of the null hypothesis of non-stationarity. The series can be considered stationary at the 5% significance level, which is a prerequisite for ARIMA modelling.

4.7.12 Differencing

```
sales_diff1 <- diff(sales_ts)
plot(sales_diff1, main = 'First Differenced Series',      ylab =
"Differenced Sales", col = "darkgreen")
```

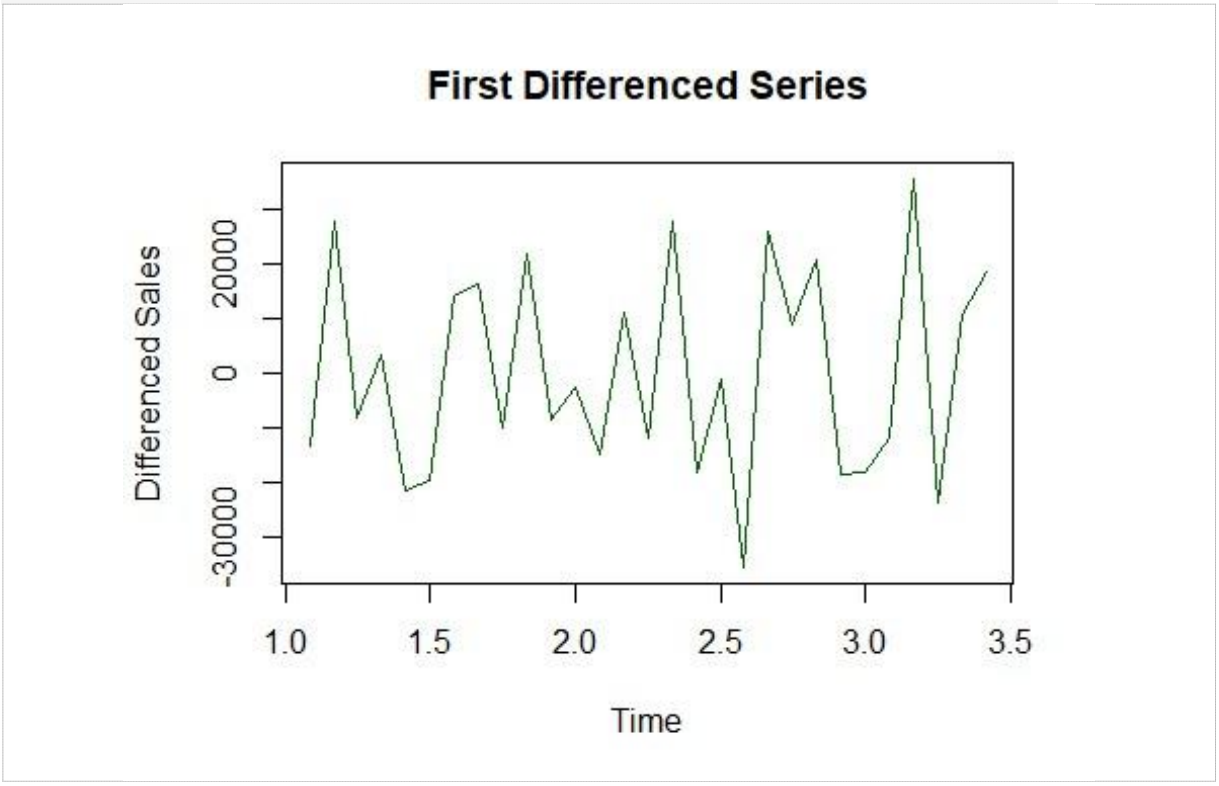


Table 4.7.9: Line Chart – First Differenced Series

First-order differencing was applied to further stabilize the series. The differenced series fluctuates around a constant mean with reduced trend patterns.

4.7.13 Autocorrelation Analysis (ACF and PACF)

```
acf(sales_ts, main = 'ACF of Monthly Retail Sales') pacf(sales_ts,
main = 'PACF of Monthly Retail Sales')
```

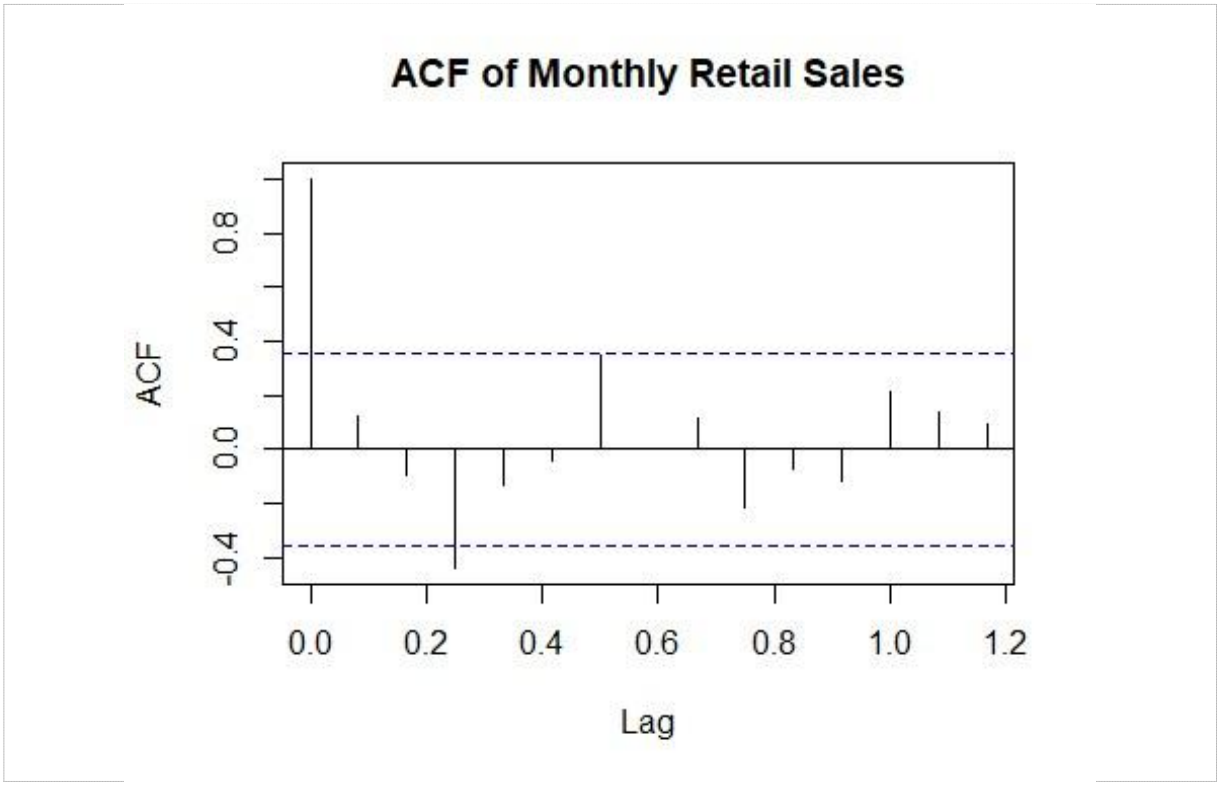


Figure 4.7.10: ACF Plot Monthly Retail Sales

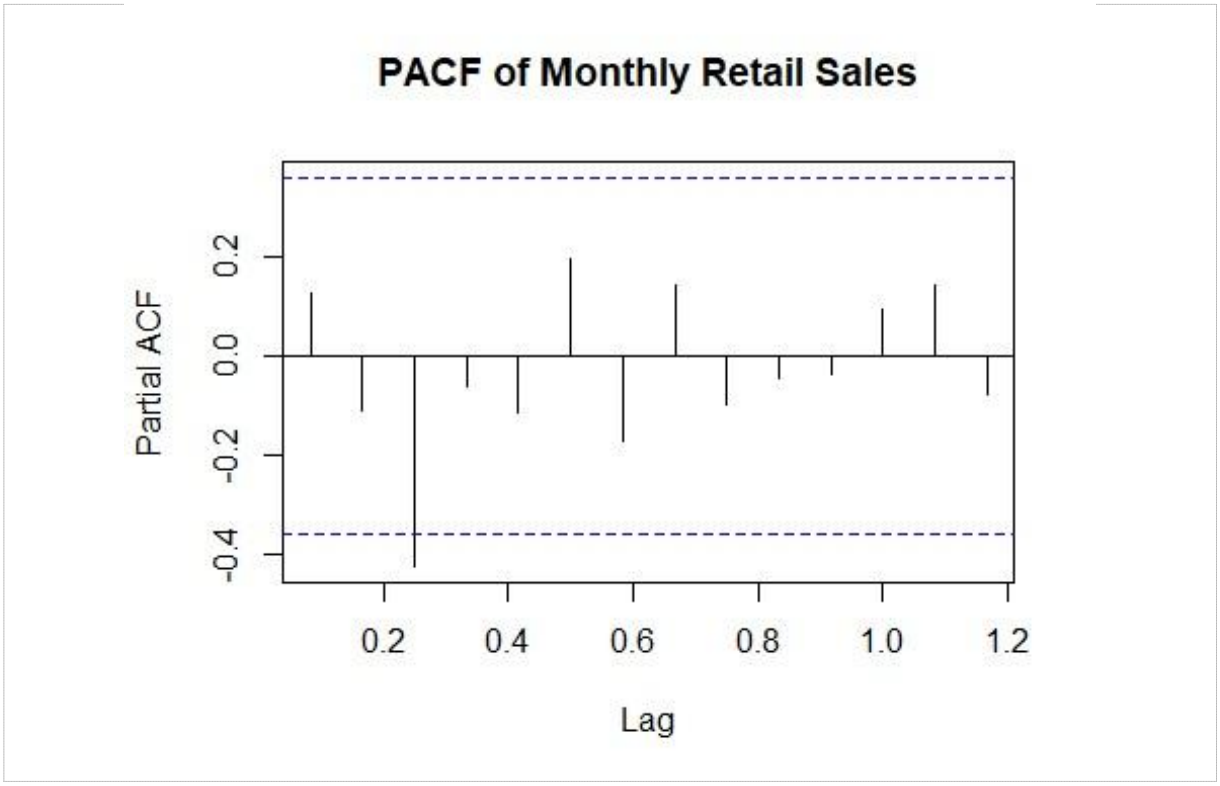


Figure 4.7.11: PACF Plot – Monthly Retail Sales

The ACF plot shows the pattern of correlations between current and lagged observations. A significant spike at lag 1 suggests an MA(1) component. The PACF plot shows significant autocorrelation at lag 1, suggesting an AR(1) component. These patterns informed the selection of ARIMA(1,1,1) as the initial model specification.

4.7.14 ARIMA Model

```
arima_model <- arima(sales_ts, order = c(1, 1, 1))
summary(arima_model)
```

```
## Coefficients: ##
ar1      ma1
## 0.1642  -1.0000
## AIC = 645.7   RMSE = 13,909   MAPE = 16.91%
```

The ARIMA(1,1,1) model: AR(1) captures dependence on previous sales, MA(1) accounts for short-term shocks. The relatively high AIC and MAPE suggest that incorporating seasonality could improve model fit.

4.7.15 SARIMA Model

```
sarima_model <- arima(sales_ts, order = c(1, 1, 1),
seasonal = list(order = c(1, 1, 1), period = 12))
summary(sarima_model)
```

```
## Coefficients: ar1=0.036, ma1=-1.000, sar1=-0.313, sma1=0.107
## AIC = 386.45   RMSE = 10,575   MAPE = 10.72%
```

The SARIMA(1,1,1)(1,1,1)[12] model shows substantially improved performance. AIC reduces from 645.7 to 386.5 and MAPE improves from 16.91% to 10.72%. Incorporating seasonal components significantly improves the model's ability to capture the retail sales dynamics.

4.7.16 Residual Diagnostics

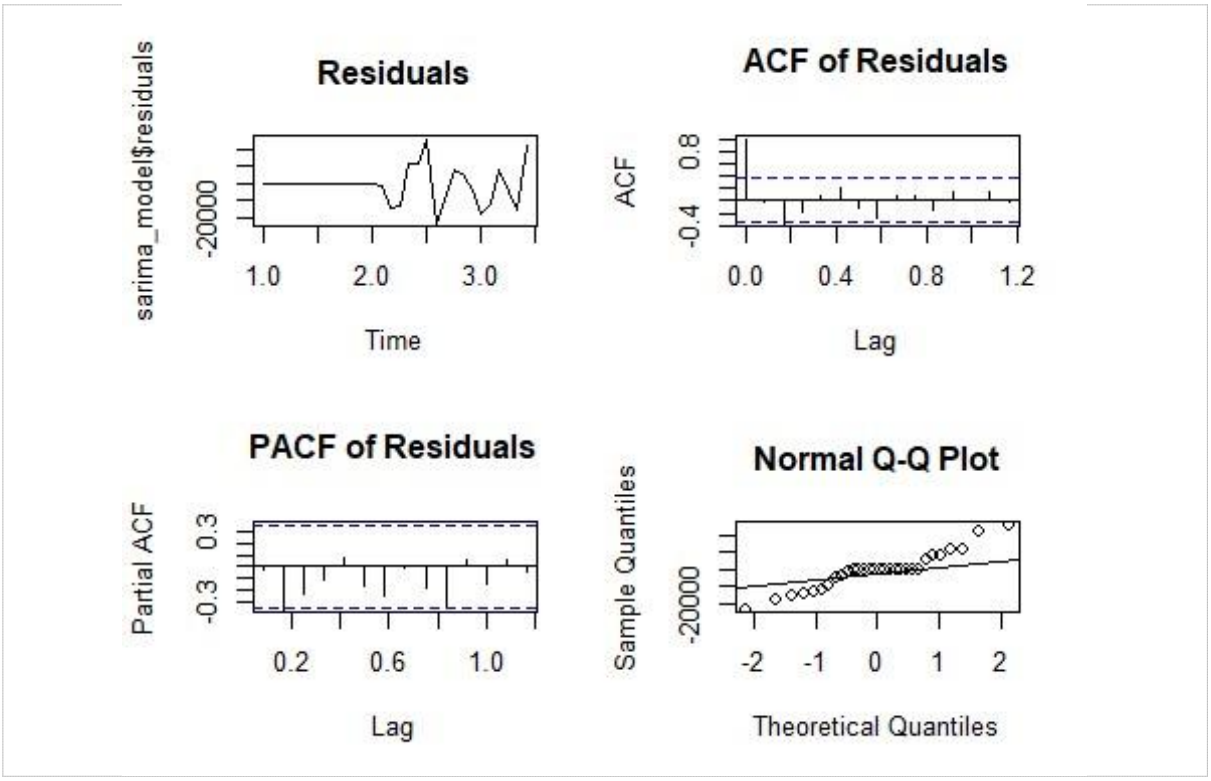


Figure 4.7.12: Residual Diagnostics – SARIMA Model (Residuals, ACF, PACF, Q-Q Plot)

Residuals fluctuate randomly around zero with no significant autocorrelation in the ACF/PACF plots. The Q-Q plot indicates approximate normality. These diagnostics confirm that the SARIMA model adequately captures the structure of the retail sales data.

4.7.17 Auto ARIMA and Sales Forecast

```
library(forecast)
model_auto <- auto.arima(sales_ts, seasonal = TRUE)
summary(model_auto)
forecast_sales <- forecast(model_auto, h = 12) plot(forecast_sales,
     main = "12-Month Sales Forecast",
ylab = "Total Monthly Sales (KES)",
xlab = "Time")
```

```
## Series: sales_ts
## ARIMA(0,0,0)(0,1,0)[12]
## AIC = 398.78   RMSE = 11,467   MAPE = 11.72%
```

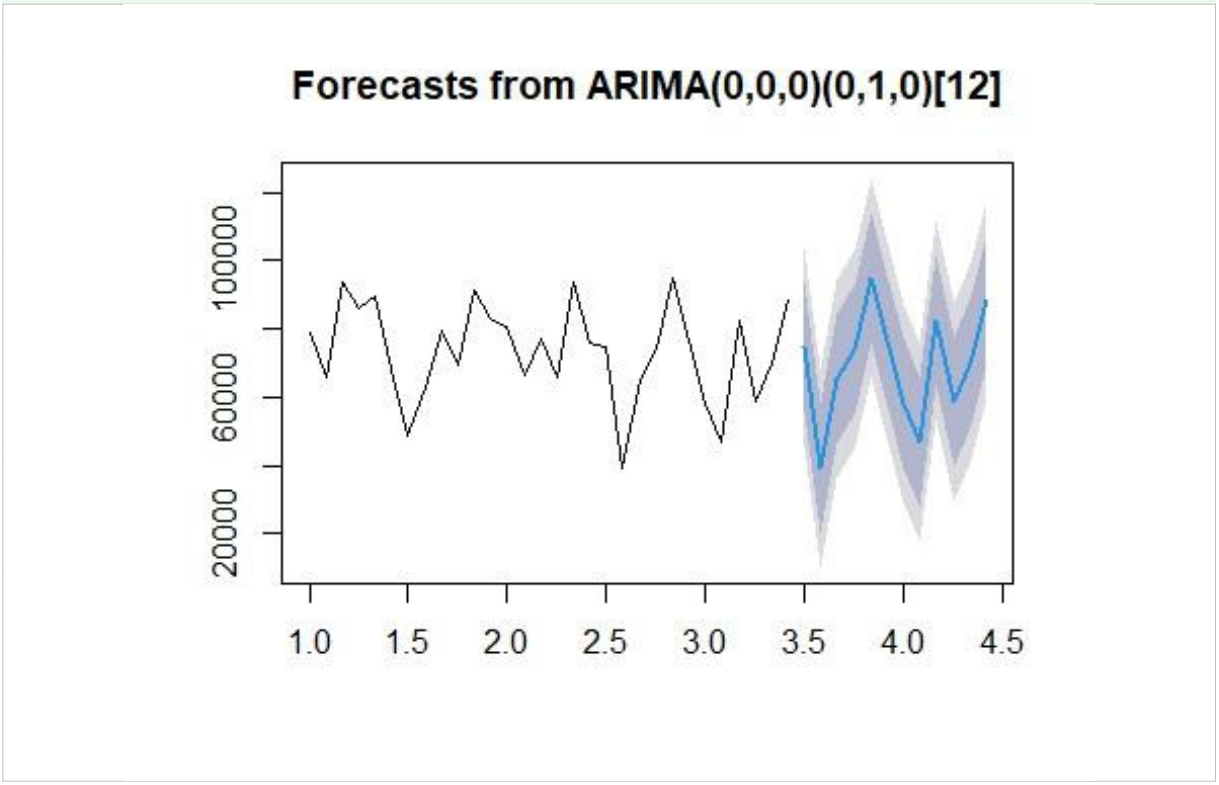


Figure 4.7.12: Forecast Plot – 12-Month Sales Forecast with Confidence Intervals

The auto ARIMA function selected ARIMA(0,0,0)(0,1,0)[12], a pure seasonal differencing model, as the optimal specification for the data. The 12-month forecast projects forward retail sales with 80% and 95% confidence intervals, providing a practical planning tool for inventory management and resource allocation. The forecast shows continuation of the observed seasonal patterns without a dominant directional trend.

4.7.18 Model Comparison Summary

Model	AIC	RMSE	MAPE(%)	Notes
ARIMA(1,1,1)	645.7	13,909	16.91	Baseline (no seasonality)
SARIMA(1,1,1)(1,1,1)[12]	386.5	10,575	10.72	Best fit lowest AIC and MAPE
Auto ARIMA(0,0,0)(0,1,0)[12]	398.8	11,467	11.72	AIC-optimal; simpler model

Table 4.7.2: Model Comparison

4.7.19 Temporal Analysis Summary

Dimension	Test	Result	Conclusion
Date (trend)	Linear regression	p = 0.241	No long-term trend
Month (seasonal)	One-way ANOVA	p = 0.006	Significant seasonal variation
Hour (intraday)	Linear regression	p = 0.683	No hourly effect
Day of Week	One-way ANOVA	p = 0.043	Significant weekly pattern
Time of Day	One-way ANOVA	p = 0.049	Morning sales highest

Table 4.7.3: Temporal analysis Summary

CHAPTER 5: DISCUSSION AND CONCLUSIONS

5.1 Summary of Findings

This project applied comprehensive statistical methods to analyze retail sales performance and payment behavior across multiple stores and time periods. The key findings across all objectives are summarized below:

General Objective

Retail sales performance is primarily driven by transaction-level factors — quantity purchased and unit price — rather than store identity, location, or payment method. Sales patterns are consistent across stores, and payment behavior is evenly distributed across five payment methods.

Comparison	F-value	p-value	Significant	Eta² (Effect)
10 Stores	0.648	0.756	No	0.003 negligible
4 Locations	0.852	0.466	No	0.001 negligible
Time of Day	3.016	0.049	Yes (borderline)	0.003 negligible
Day of Week	2.177	0.043	Yes	0.003 negligible
Monthly (30 groups)	1.800	0.006	Yes	0.022 small

Specific Objective 1: Descriptive Analysis

The mean transaction value is KES 1,104 (median KES 839), confirming a right-skewed distribution. Monitors and Chairs generate the highest average revenues. The Shapiro-Wilk test confirms significant non-normality, justifying Box-Cox transformation in subsequent regression analyses.

Specific Objective 2: Payment and Product Association

The Chi-square test revealed no statistically significant association between payment method and product category ($X^2 = 21.008$, df = 24, $p = 0.638$). Cramer's V = 0.051 confirms a negligible effect. Customers' payment choices are entirely independent of the product they purchase.

Specific Objective 3: Sales Performance Comparison

ANOVA found no statistically significant differences in mean sales across stores ($F = 0.648$, $p = 0.756$) or locations ($F = 0.852$, $p = 0.466$). A borderline significant difference was detected across time of day ($F = 3.016$, $p = 0.049$), with morning transactions generating the highest average sales (KES 1,164). Effect sizes were negligible throughout ($Eta^2 < 0.003$).

Specific Objective 4: Regression Modelling

The interaction regression model (R^2 = 0.936) demonstrated that Quantity and Unit price are the dominant determinants of total sales value, with a significant interaction effect (F = 615.38, p < 0.001). Product type, payment method, and time of day do not predict sales when controlling for quantity and price. Logistic regression achieved 90.35% accuracy in classifying high-value transactions.

Simple Linear (Quantity)	**R^2 = 0.384**	**—**	**Quantity significant (β=1.583, p<0.001)**
Multiple Linear	R^2 = 0.916	8,722.8	Quantity + Unit Price only (p<0.001)
Interaction Model ★ Best	R^2 = 0.936	8,184.5	Quantity × Unit price interaction (F=615.38, p<0.001)
Logistic Regression	90.35%, AUC≈0.98+	782.7	Qty OR=4.88, Unit price OR=1.043

Specific Objective 5: Temporal Analysis

Monthly seasonality significantly influences sales (p = 0.006), as does day of week (p = 0.043). No long-term chronological trend was detected. The SARIMA(1,1,1)(1,1,1)[12] model provides the best time series fit (AIC = 386.5, MAPE = 10.72%).

Dimension	**Test**	**p-value**	**Conclusion**
Long-term trend	Linear regression	0.241	No directional trend 2023–2025
Monthly seasonality	One-way ANOVA	0.006	Significant — February peaks, July lowest
Hourly (intraday)	Linear regression	0.683	No hourly pattern — uniform throughout day
Day of Week	One-way ANOVA	0.043	Tuesday & Thursday peak; Monday lowest
Time of Day	One-way ANOVA	0.049	Morning highest; Afternoon lowest

5.2 Practical Implications

- Pricing and Quantity Strategy: Retailers should focus on upselling and Cross selling strategies. Higher-priced items combined with bulk purchases generate disproportionately higher revenue due to the interaction effect.
- Staffing and Operations: Morning transactions consistently generate the most revenue. Retailers should ensure full staffing and complete product availability during morning hours.

- Mid-Week Promotions: Tuesday and Thursday are peak sales days. Targeted promotions on lower-performing days (Monday, Friday) could help distribute demand more evenly throughout the week.
- Seasonal Planning: Monthly seasonality calls for proactive seasonal inventory management, with higher stock levels in historically high-performing months and targeted marketing campaigns.
- Payment Infrastructure: The balanced payment distribution confirms that maintaining all five payment options is essential. No single method should be incentivized or discouraged.
- Store Performance Evaluation: The uniform performance across stores suggests centralized management is effective. Individual stores do not require differentiated strategies based on performance alone.

5.3 Conclusions

This project successfully achieved all stated objectives through rigorous application of descriptive statistics, chi-square testing, ANOVA, regression modelling (simple, multiple, interaction, and logistic), and time series analysis (ARIMA, SARIMA, and decomposition).

The overarching conclusion is that retail sales performance in this dataset is governed primarily by transaction-level economics — how much customers buy and at what price — rather than by where, when, or how they pay. This finding simplifies the management challenge: revenue optimization should focus on product pricing strategy and quantity incentives (bundles, volume discounts) rather than store-specific or time specific interventions.

The analytical framework developed in this project — combining data cleaning, visualization, hypothesis testing, regression, and time series forecasting — provides a replicable, scalable model for continuous retail performance monitoring and evidence-based decision-making.

5.4 RECOMMENDATIONS

5.4.1 Practical Recommendations for Retail Management

	Recommendation	Statistical Basis
1	Implement volume discount pricing — the Quantity × Unit price interaction is the strongest revenue driver. Structured bundle deals (e.g., buy 5 items, get 10% off) will maximize revenue especially for high-priced Monitor and Chair products.	Interaction regression: F=615.38, p<0.001; Interaction beta=0.003
2	Prioritize morning staffing and full product availability — morning transactions consistently generate the highest average revenue. Ensure complete staffing and stocked shelves before 12:00 noon.	Time of Day ANOVA: p=0.049; Morning mean = KES 1,164 vs Afternoon KES 1,043
3	Launch targeted mid-week promotions — Tuesday and Thursday are peak sales days. Introduce promotions on Monday (lowest day, KES 978 mean) and Friday to distribute demand more evenly across the week.	Day of Week ANOVA: p=0.043; Monday lowest, Tuesday highest (KES 1,192)
4	Build seasonal inventory plans using the SARIMA forecast — February peaks and July troughs are statistically confirmed. Stock Monitor and Chair categories ahead of peak months.	Monthly ANOVA: p=0.006; SARIMA MAPE=10.72%; Feb peak=KES 1,293
5	Maintain all five payment options equally — no payment method is associated with product type or transaction value. Eliminating or discouraging any payment option would reduce customer convenience without revenue benefit.	Chi-square: p=0.638, Cramer's V=0.051 — complete independence
6	Investigate Store S2 qualitatively — while ANOVA shows no statistically significant store-level differences, Store S2 consistently records the lowest mean transaction value (KES 1,006 vs network mean KES 1,104). A qualitative review of its product mix and pricing is warranted.	Store ANOVA: p=0.756; S2 lowest mean across all stores
7	Deploy the logistic model for proactive upselling identifies high-value transaction likelihood from Quantity and Unit price alone with 90.35% accuracy. Train cashiers to recognize high-potential customers and suggest complementary products.	Logistic regression: accuracy=90.35%, AUC>0.98
8	Introduce quantity-based loyalty rewards — each additional unit purchased multiplies the odds of a highvalue transaction by 4.88. Loyalty programs rewarding bulk purchases represent a high-ROI customer retention strategy.	Logistic regression: Quantity OR=4.88, p<0.001

Table 5.4.1: Recommendations for Retail Management.

5.4.2 Recommendations for Future Statistical Research

No.	Recommendation	Gap Addressed
1	Apply Two-Way ANOVA or MANOVA to simultaneously test joint effects of Store and Time of Day on sales. Current analysis tests each factor independently, missing potential interaction effects between operational variables.	Joint factor testing Objective 3
2	Use mixed-effects (multilevel) regression models that account for the nested structure of transactions within stores, cashiers, and managers. These models produce more precise standard errors and control for clustering effects.	Hierarchical data structure Objectives 1, 3, 4
3	Apply machine learning models — Random Forest, Gradient Boosting, and Support Vector Machines — and compare their predictive accuracy against the classical regression $R^2 = 0.936$ benchmark established in this project.	Predictive modelling benchmark Objective 4
4	Conduct seasonal decomposition using STL (Seasonal Trend decomposition using LOESS) as a more robust alternative to classical additive decomposition, particularly when seasonal variance is non-constant across years.	Robust time series decomposition Objective 5
5	Expand the dataset to include customer demographics (age, loyalty tier) and external variables (public holidays, local events, competitor promotions). Use SARIMAX (SARIMA with exogenous regressors) to test whether these external factors improve forecast accuracy.	External variables in forecasting Objective 5
6	Apply survival analysis or hazard models to study time-to repeat-purchase behavior if customer-level transaction histories become available. This would extend the project from transaction-level to customer-level analytics.	Customer retention and repeat purchase dynamics

Table 5.4.1: Recommendations for Future Statistical Research.

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APPENDICES

```
kable(summary(select(Retail_data, Quantity, UnitPrice, TotalPrice, Hour,
Highvalue)),       caption = "Summary of Numeric Variables")
```

```
pareto_data <- Retail_data %>%   group_by(Product)
%>%   summarise(TotalRevenue = sum(TotalPrice)) %>%
arrange(desc(TotalRevenue)) %>%   mutate(
    Pct     = TotalRevenue / sum(TotalRevenue) * 100,
    CumPct = cumsum(Pct),
    Product = factor(Product, levels = Product)   )

# Pareto Chart
ggplot(pareto_data, aes(x = Product)) +
  geom_col(aes(y = Pct), fill = "steelblue", alpha = 0.8) +
  geom_line(aes(y = CumPct, group = 1), colour = "red", linewidth = 1) +
geom_point(aes(y = CumPct), colour = "red", size = 2.5) +
  geom_hline(yintercept  =  80,  linetype  =  "dashed",  colour  =  "darkgreen")  +
labs(title = "Pareto Chart: Revenue Contribution by Product",        x = "Product",
y = "Revenue Share (%)") +   theme_minimal()
```

```
par(mfrow = c(2, 2))
plot(sarima_model$residuals, main = 'Residuals') acf(sarima_model$residuals,
main = 'ACF of Residuals') pacf(sarima_model$residuals, main = 'PACF of
Residuals') qqnorm(sarima_model$residuals)
```